

Magnetism of massive (and other) stars:

Recent results and open questions

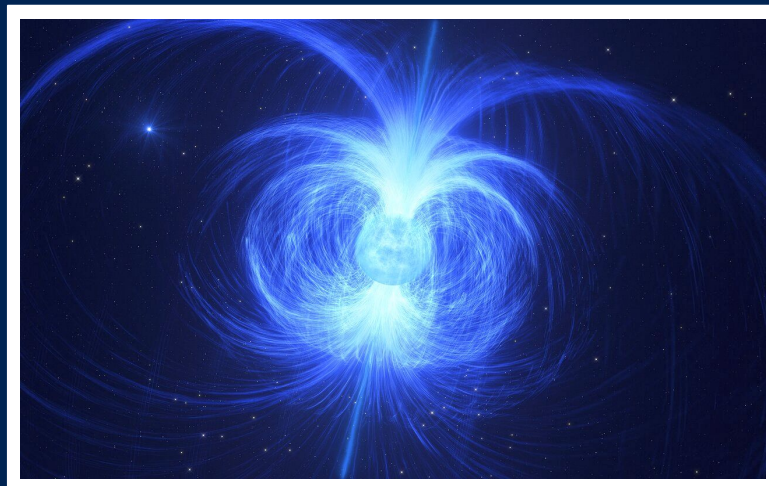
James Barron and Gregg Wade

Queen's University and RMC, Canada

BRAHMS 2026

August 18, 2026

HD 45166



Credit: ESO/L. Calçada



Queen's
UNIVERSITY

Outline

1. Introduction
2. Magnetic fields of hot stars near current detection limits.
3. Magnetic fields and binary evolution.
4. Magnetic fields of evolved stars.

1. Introduction

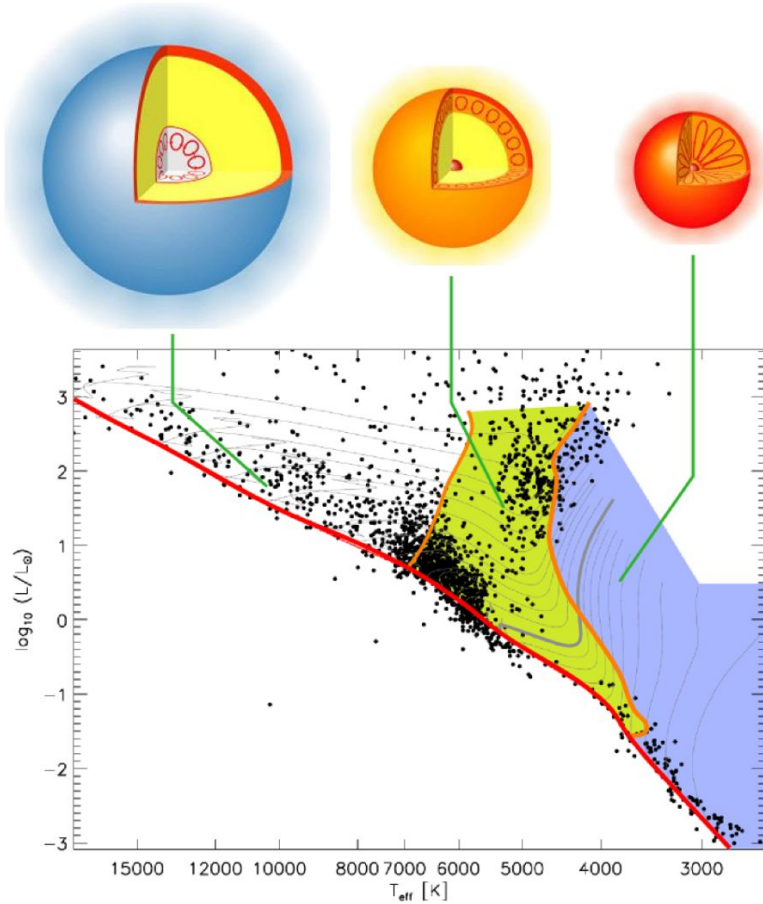
Objectives

1. Understanding magnetic fields as key physical ingredients in theories of stellar structure and evolution (what you do for us!)
2. Identifying ways that magnetic fields can explain phenomena you observe (what we do for you!)

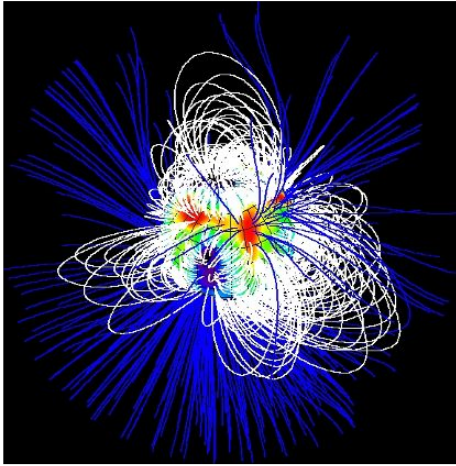
What magnetic fields can do

- Introducing or suppressing differential rotation
- Modifying angular momentum transport
- Enhancing or suppressing chemical mixing
- Modifying bulk rotation
- Quenching bulk mass loss
- Introducing surface and wind structures
- Tracing dynamical processes
- ...

Key contextual framework - convection and magnetism



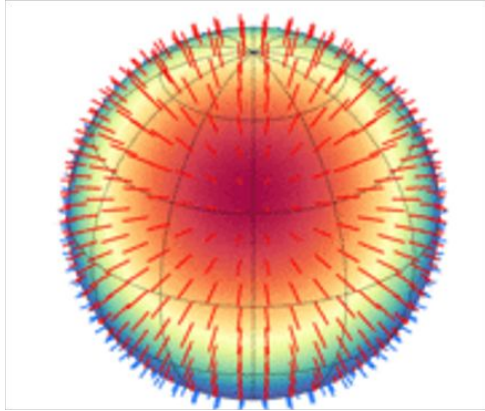
Dynamo magnetic fields



SU Aur,
Wikimedia
commons

- Magnetic fields attributed to **dynamos** are **ubiquitous** among **cool stars**
- Fields often show **large contrast** of **local** (strong, kG) versus **global** (weak, G) **components**
- **Global field strengths** are strongly correlated with rotation and show clear systematic changes with mass
- **Field topologies** can be stable, or exhibit (sometimes quasi-periodic) secular evolution

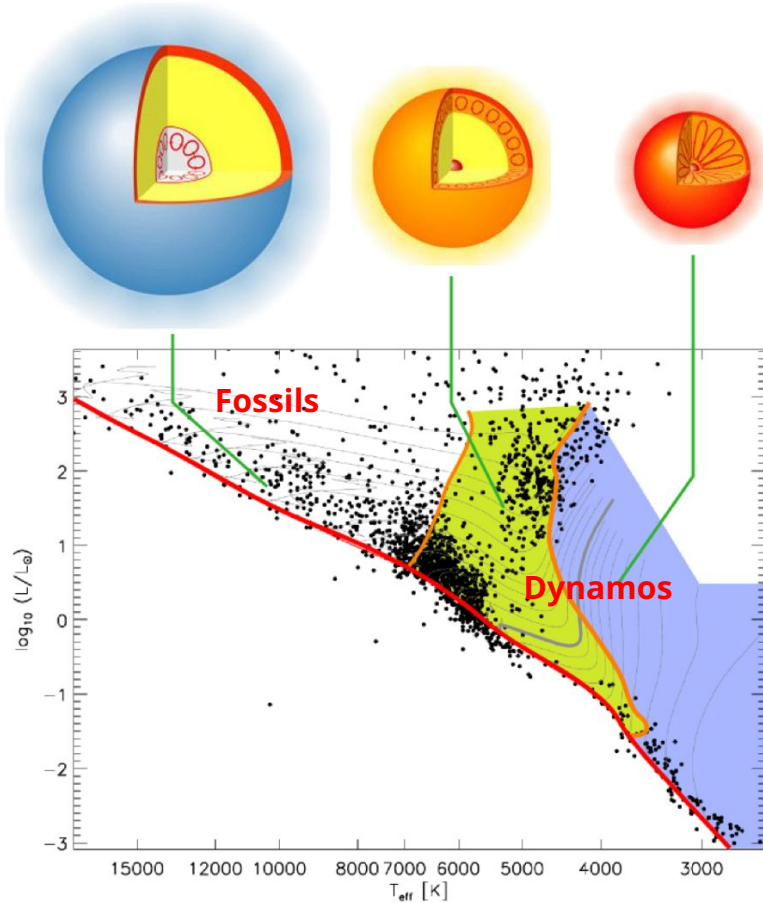
Fossil magnetic fields



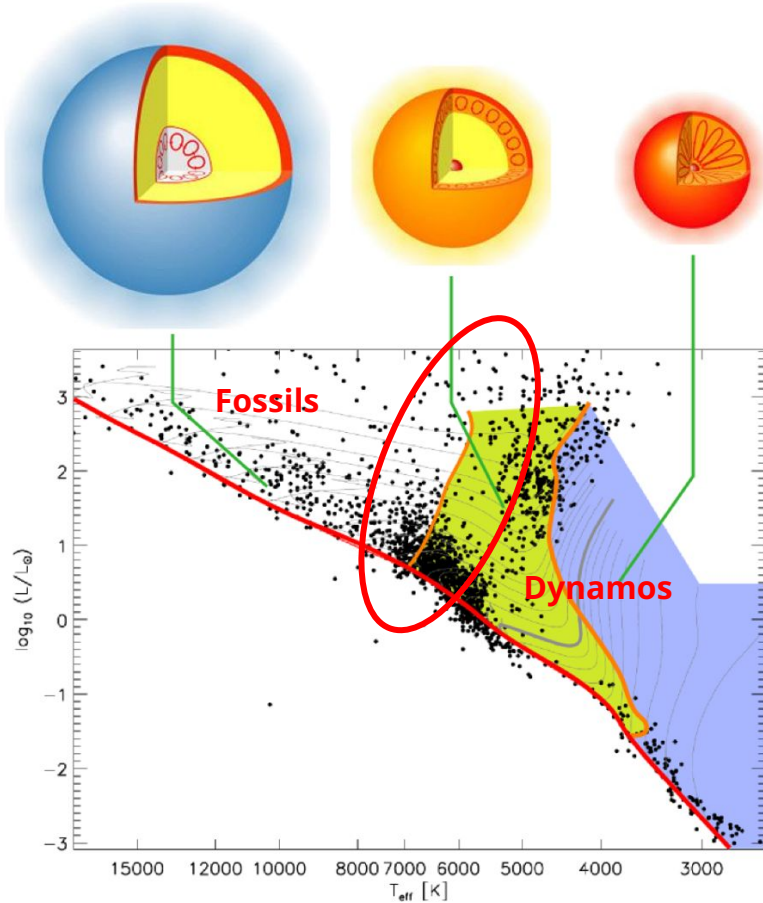
Courtesy O.
Kochukhov,
U. Uppsala

- ~10% of MS A, B and O stars exhibit global, organized magnetic fields
- Most are approximately oblique dipoles with $B \sim 100$ to 10,000 G
- Local field contrast is low
- Their field strengths are largely uncorrelated with rotation and show no systematic change with mass
- Their field topologies are very stable
- Some “non-magnetic” A stars have very weak fields that may be “failed fossils”

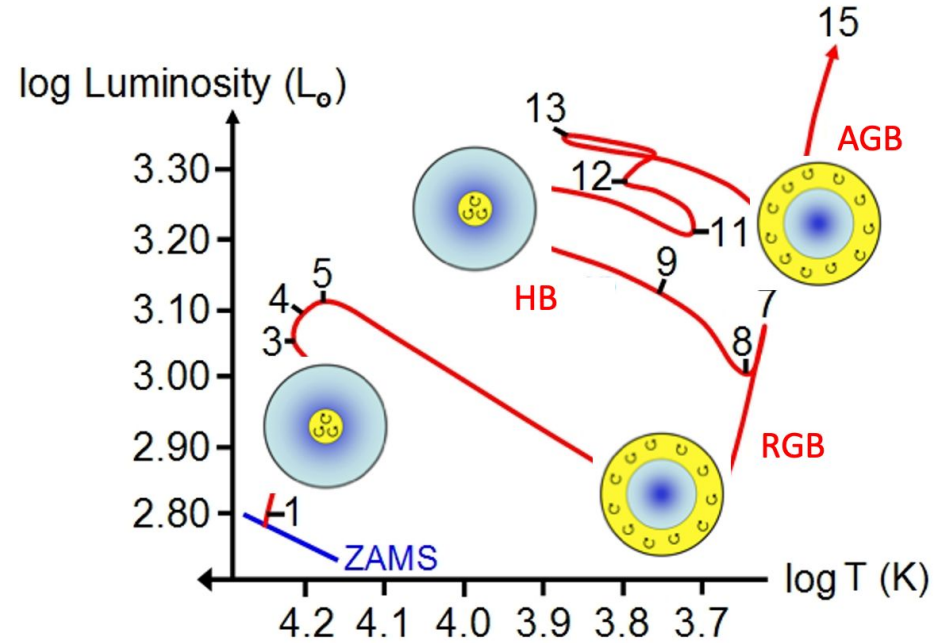
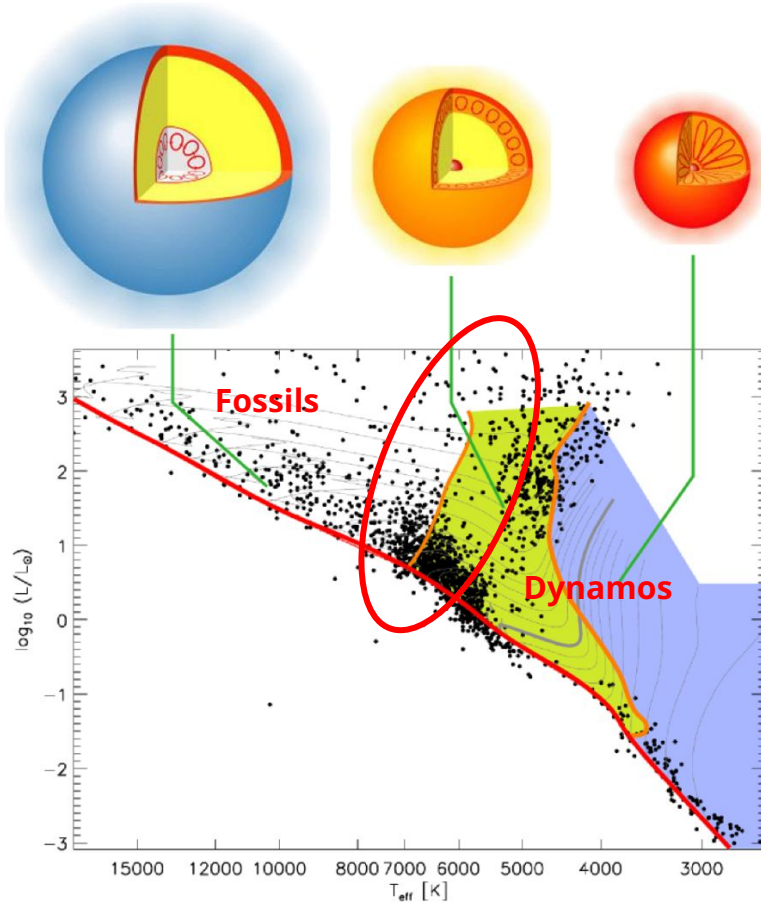
Key contextual framework - surface magnetic fields



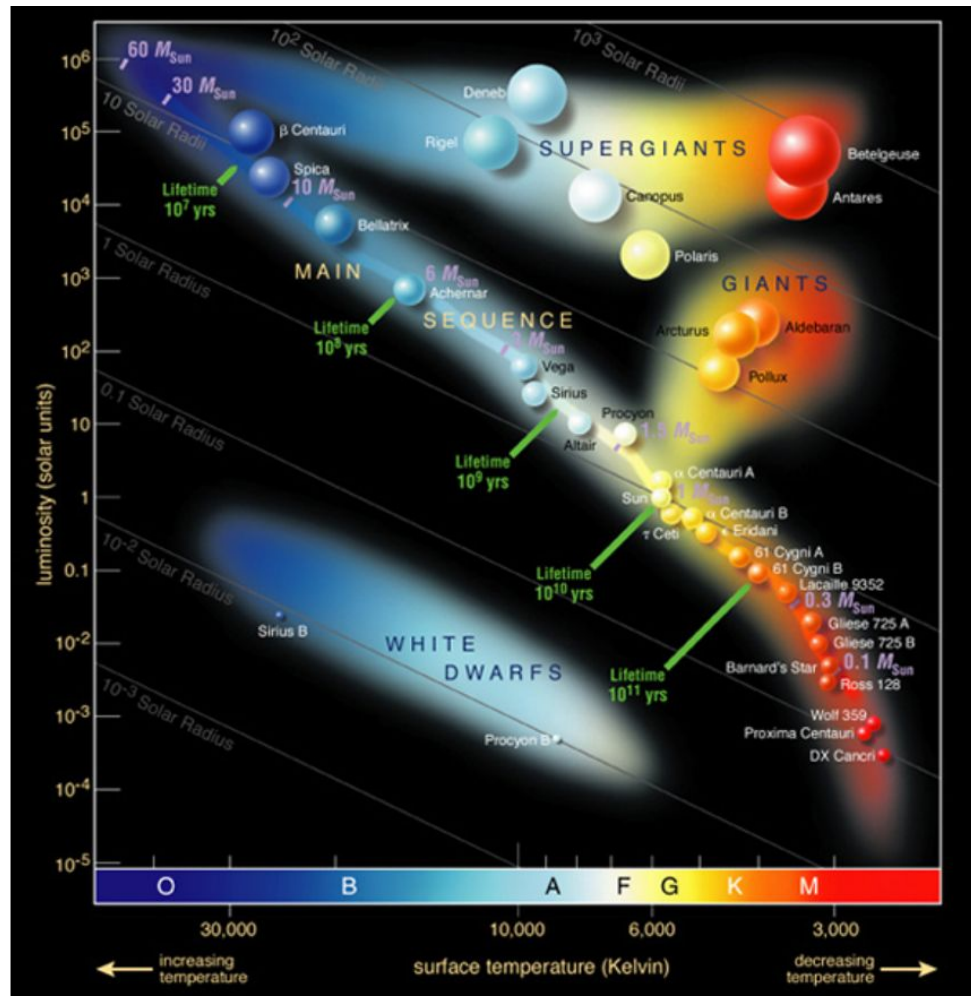
Key contextual framework - surface magnetic fields



Key contextual framework - interior magnetic fields

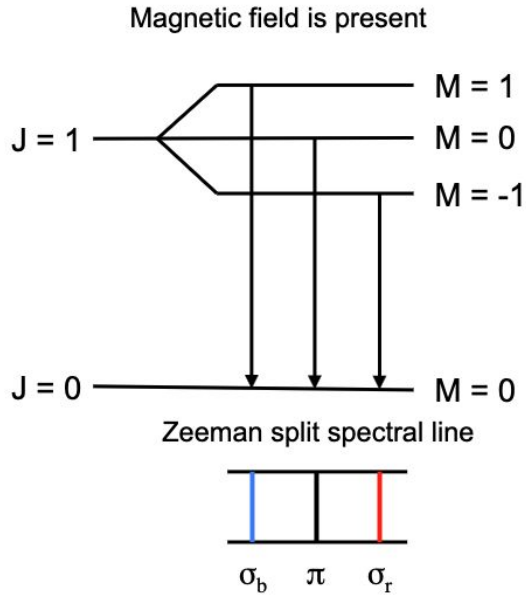


Stellar
magnetic
fields have
been
measured
across the
HRD...

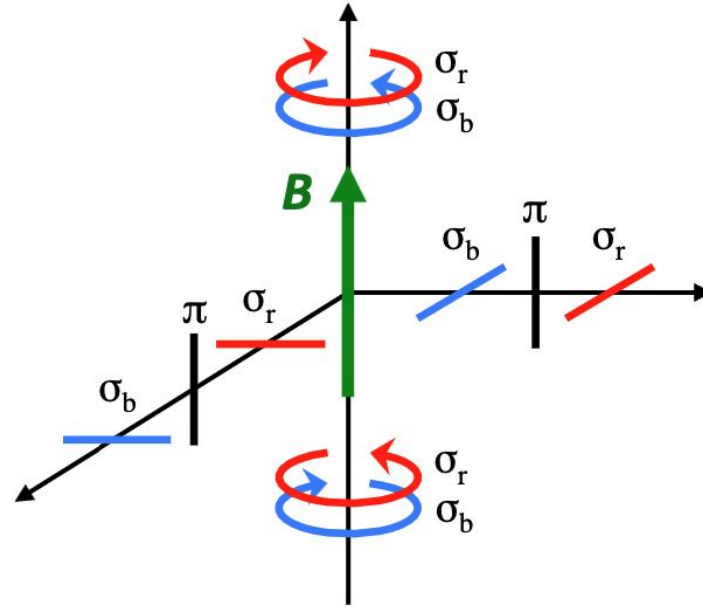


Spectropolarimetry and Magnetic Fields

Zeeman Line Splitting

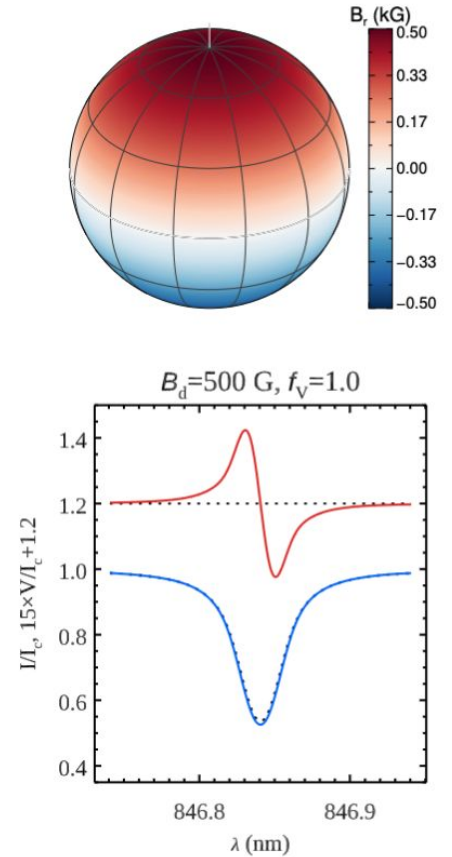


Magnetic Induced Polarization



Kochukhov 2021

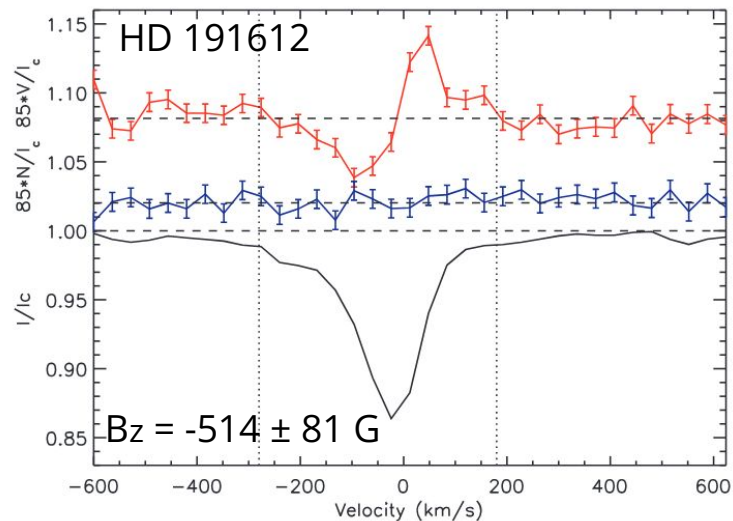
Magnetic Field Geometry



$$I = \text{star symbol} \quad V = \text{circle with arrow} - \text{circle with arrow}$$

Least Squares Deconvolution and Longitudinal Magnetic Fields

LSD cross-correlation technique

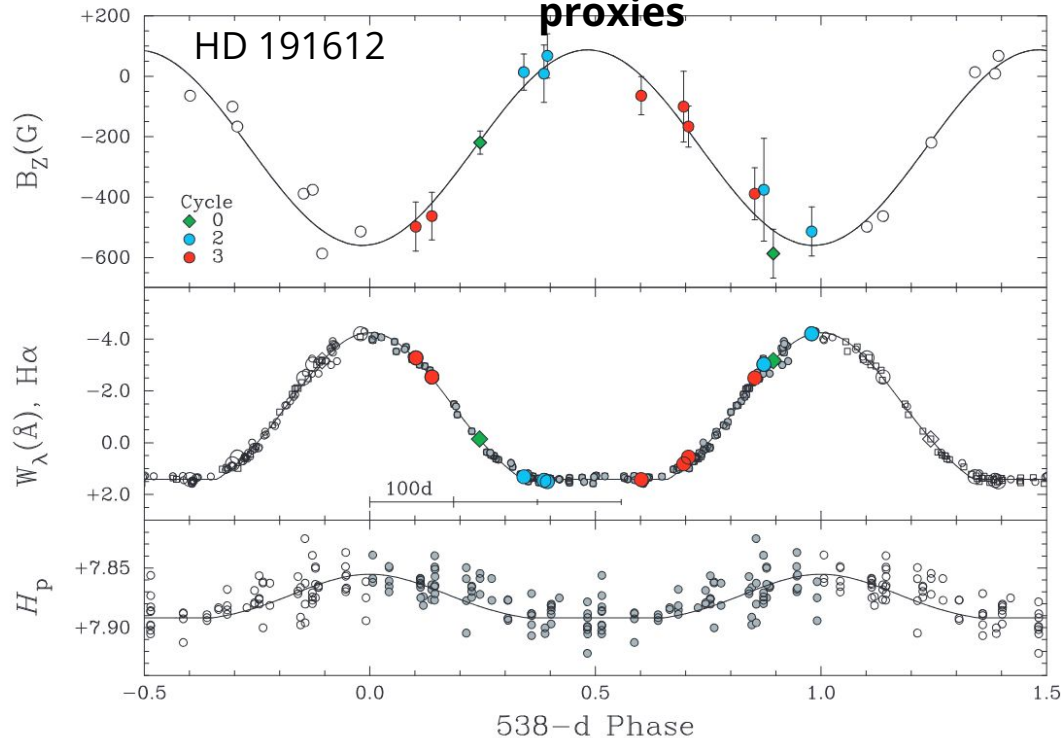


B_z Measurement

$$B_{\ell} = -2.14 \times 10^{11} \frac{\int v V(v) dv}{\lambda z c \int [I_c - I(v)] dv}$$

The line-of-sight component of the magnetic field averaged over the visible stellar disk.

B_z curves trace rotation and correlate with magnetic proxies



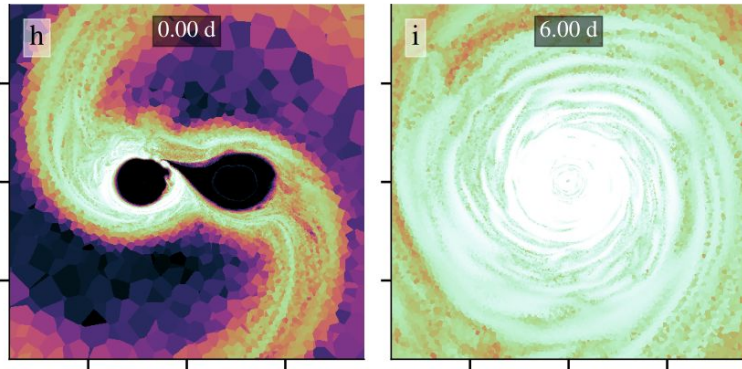
2. Magnetic fields of hot stars near current detection limits

Detected surface magnetic fields in OBA stars are relatively rare (<10%), typically strong (~100 G - 10 kG) and stable over decades long observations. Weak surface magnetic fields are observed in a subset of A-type stars forming a bimodal *B* field distribution.

1. What sets the distribution of OBA star surface magnetic fields?
2. How do fossil fields evolve over MS lifetimes?
3. Does the A star 'magnetic desert' extend to higher masses?
4. How weak are the smallest detectable magnetic fields in OB stars?
5. What are the physical and observational consequences of such weak fields?

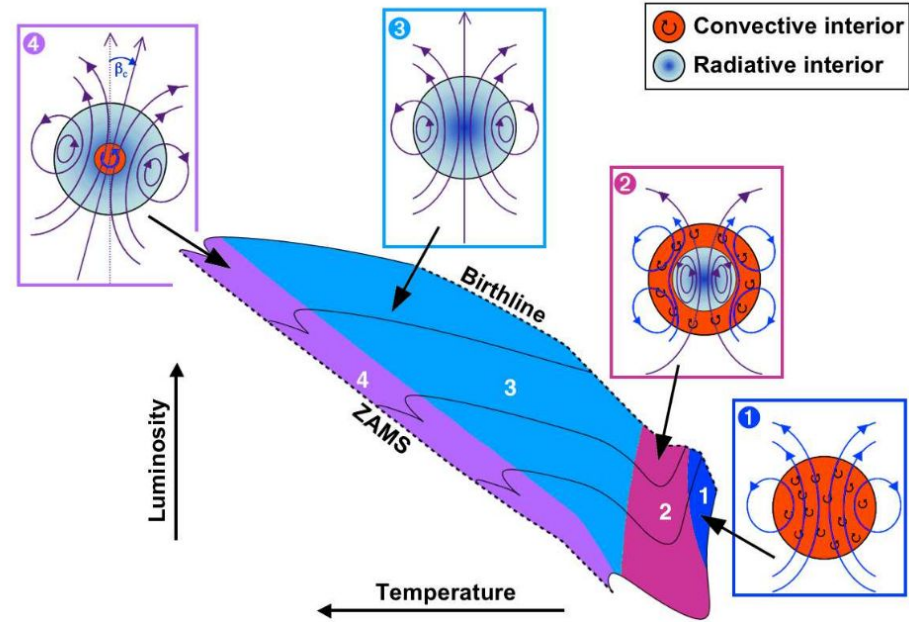
How to generate a fossil magnetic field?

Stellar Mergers and Mass Transfer



Schneider+ 2019

Dynamo Amplification During PMS

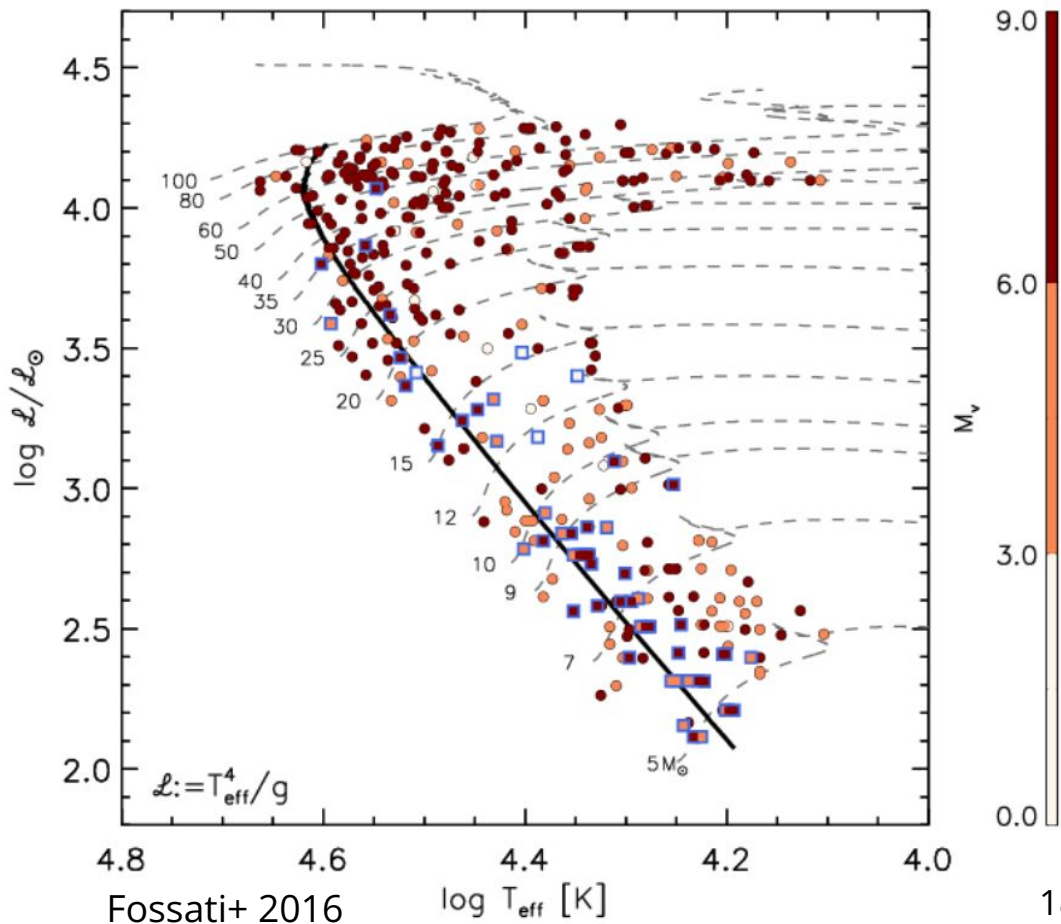


Neiner+ 2015

Magnetic detections in O and B stars are rarer near the TAMS

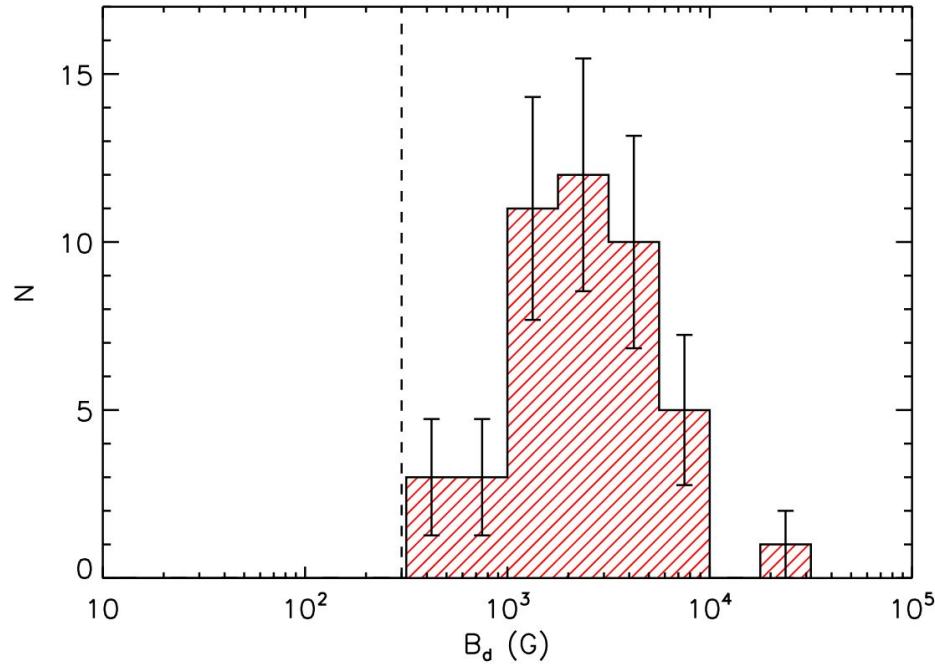
Possible Explanations

- Ohmic decay.
- Field destruction due to instability:
 - Differential rotation.
 - Subsurface convection (Jermyn & Cantiello 2020).
- Flux conservation and increasing radius.
$$B_{surf}(t) = B_{surf}(t_{ZAMS}) \left(\frac{R(t_{ZAMS})}{R(t)} \right)^2.$$
- Observational biases: field strength, rotation rate, number of photospheric spectral lines, intrinsic variability, binarity.



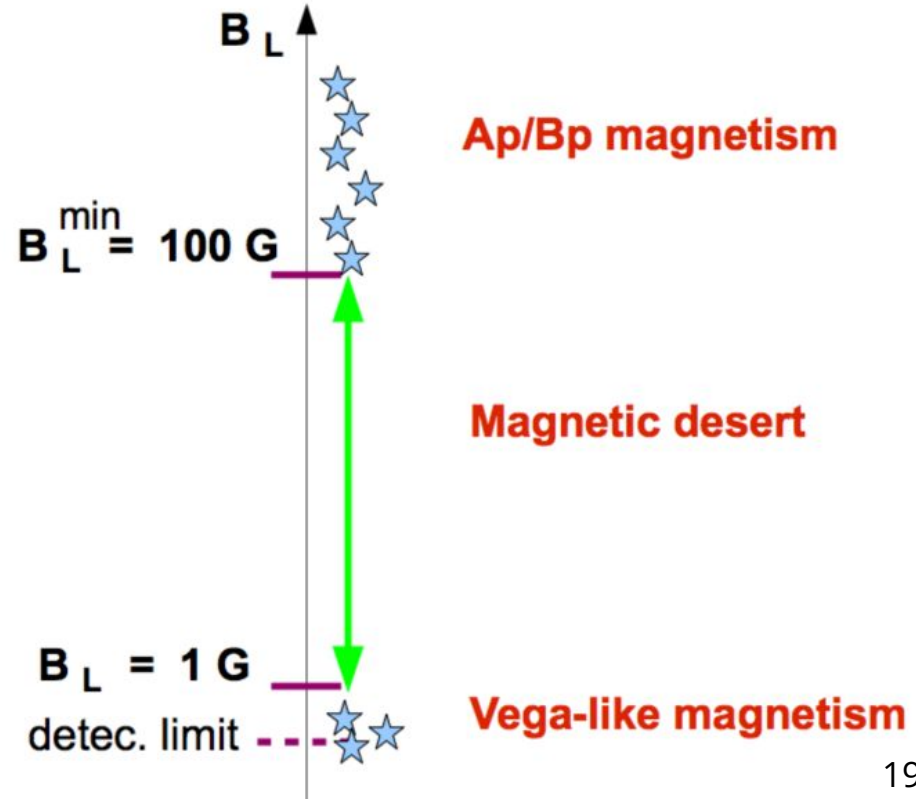
How far does the bimodal distribution extend?

Ap/Bp Star 100 pc Sample



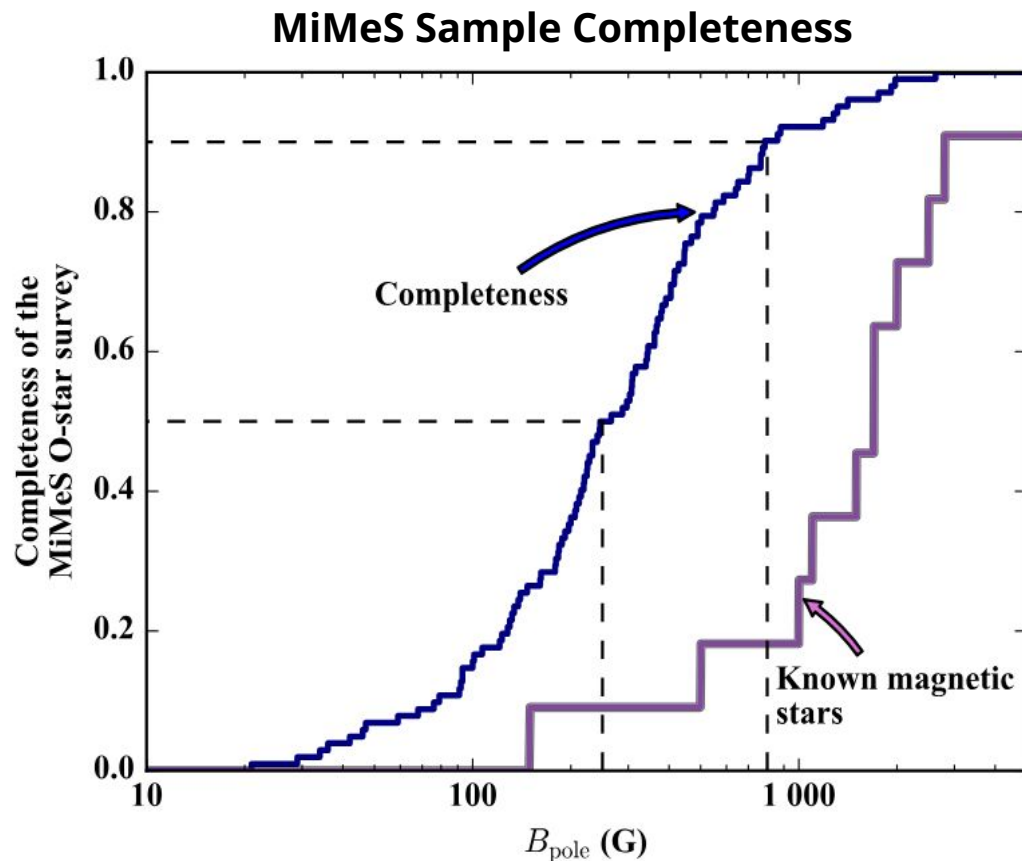
Sikora+ 2019

Magnetic Desert



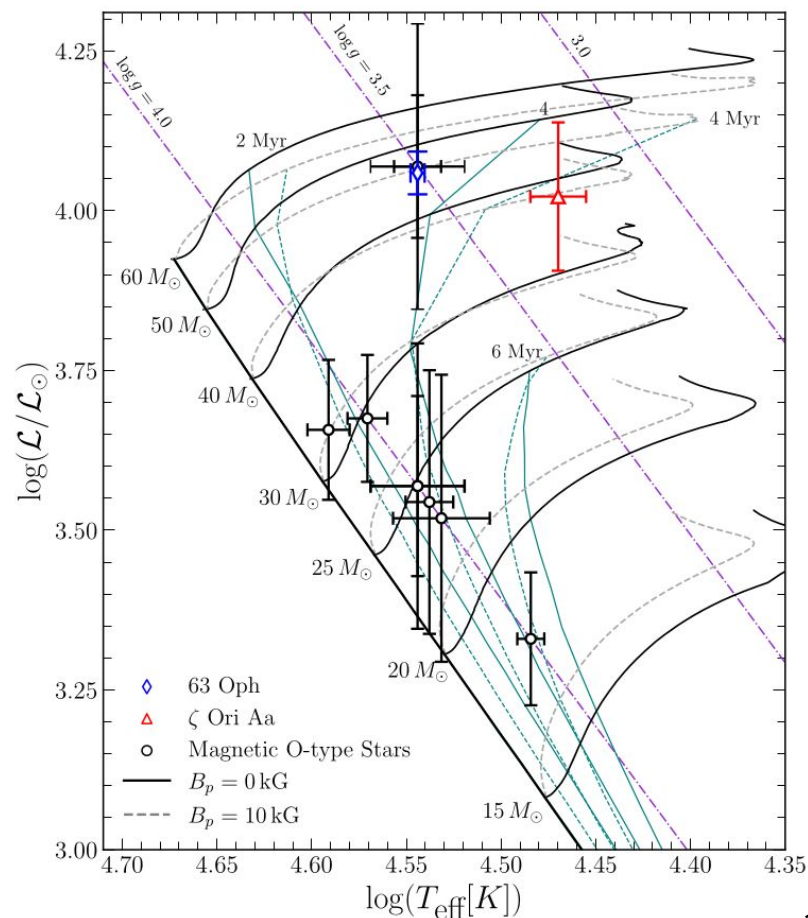
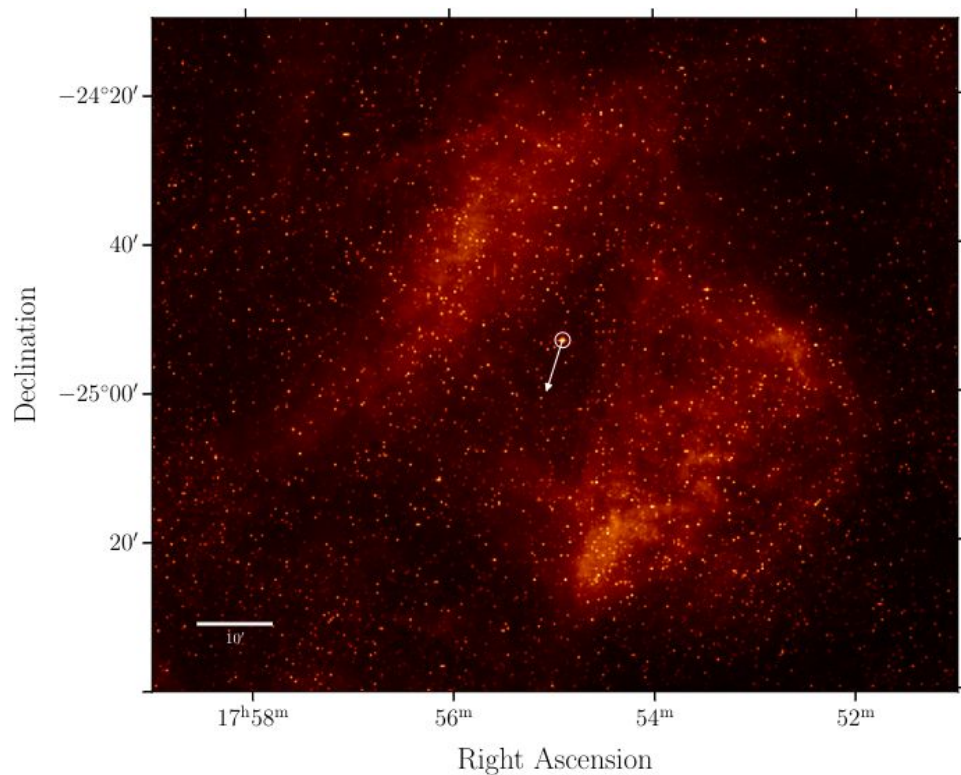
Are we missing weak field O-type stars?

- Bayesian analysis of the MiMeS O star sample consisting of 432 Stokes V observations of 97 O-type stars.
- ~12 Confident O star magnetic detections.
- Rule out all non-detected stars hosting a dipolar magnetic field with strength just below detection limit.
- Cannot confirm or refute magnetic flux decay in O-type stars.
- Initial Magnetic Field function (IBF) likely bimodal - young O stars have weak/absent fields or strong fields.



63 Oph: A transitional magnetic O-type star?

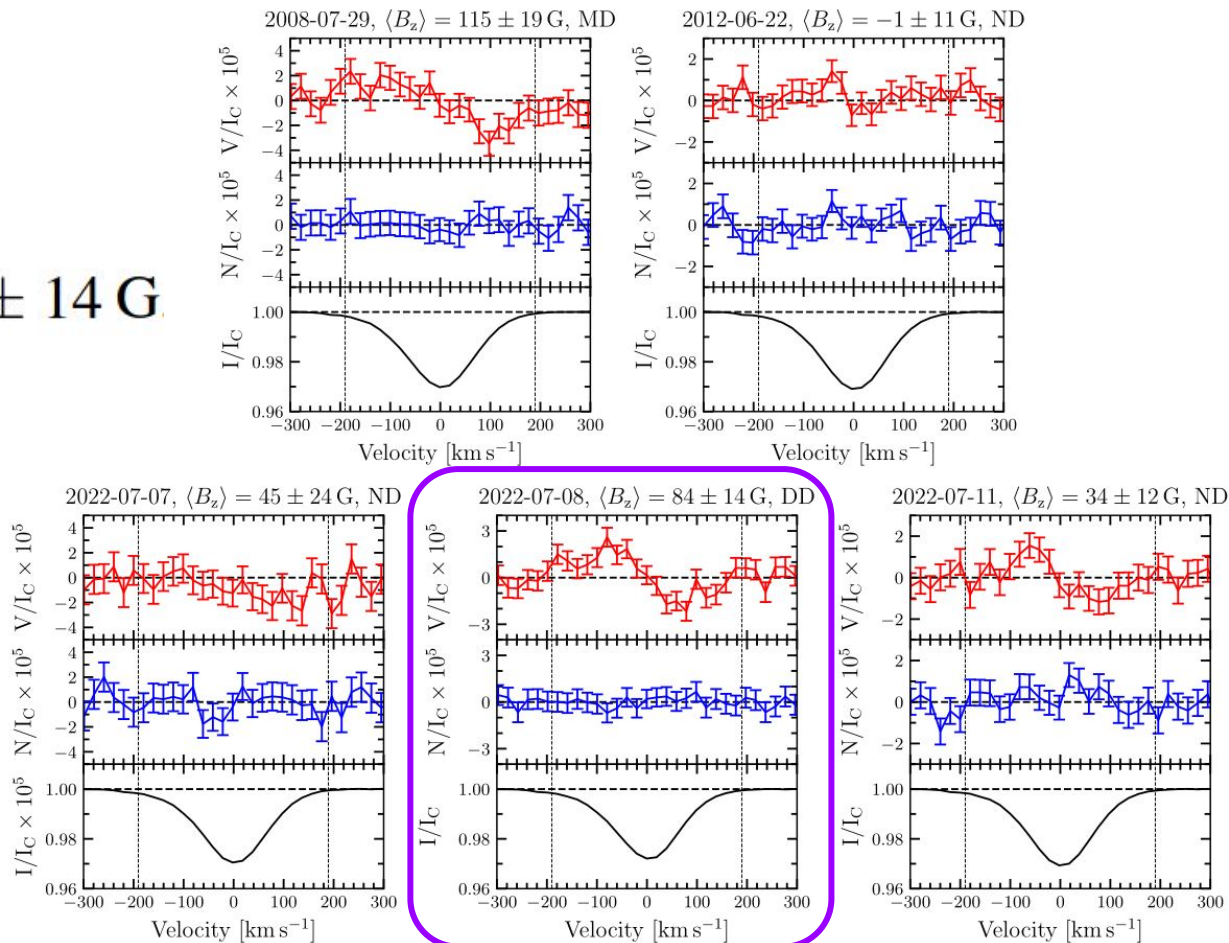
- O8 II((f))
- $T_{\text{eff}} = \sim 35\,000\text{ K}$
- $\log g \sim 3.5$
- $M_{\text{evol}} \sim 48\,M_{\odot}$
- $R_{\text{evol}} \sim 20\,R_{\odot}$
- $d \sim 1\text{ kpc}$



63 Oph: Dipole field strength > 300 G

Barron+ 2025

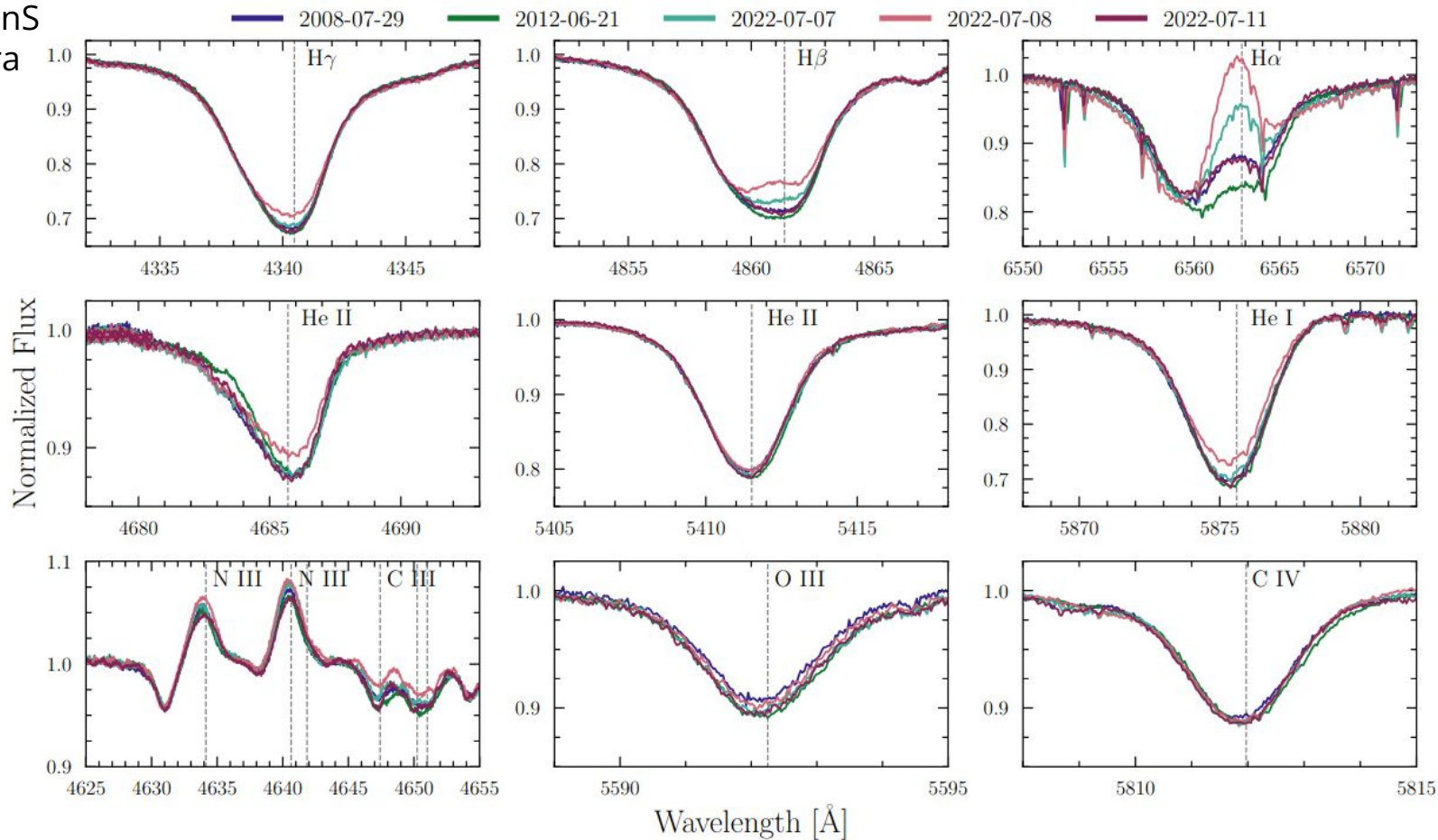
$$\langle B_z \rangle = 84 \pm 14 \text{ G}$$



63 Oph: Line Profile Variability

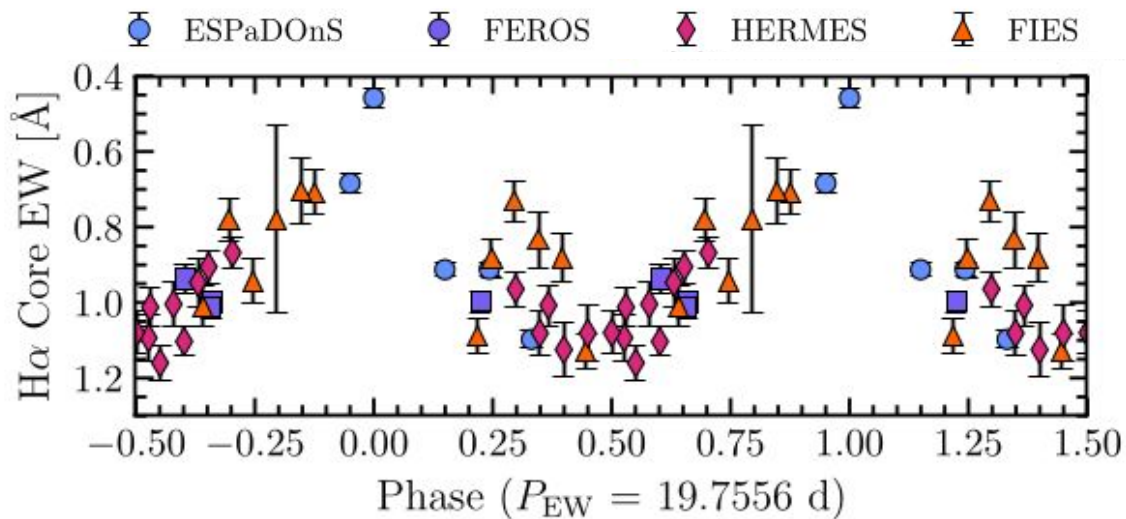
Barron+ 2025

ESPaDOnS
Spectra

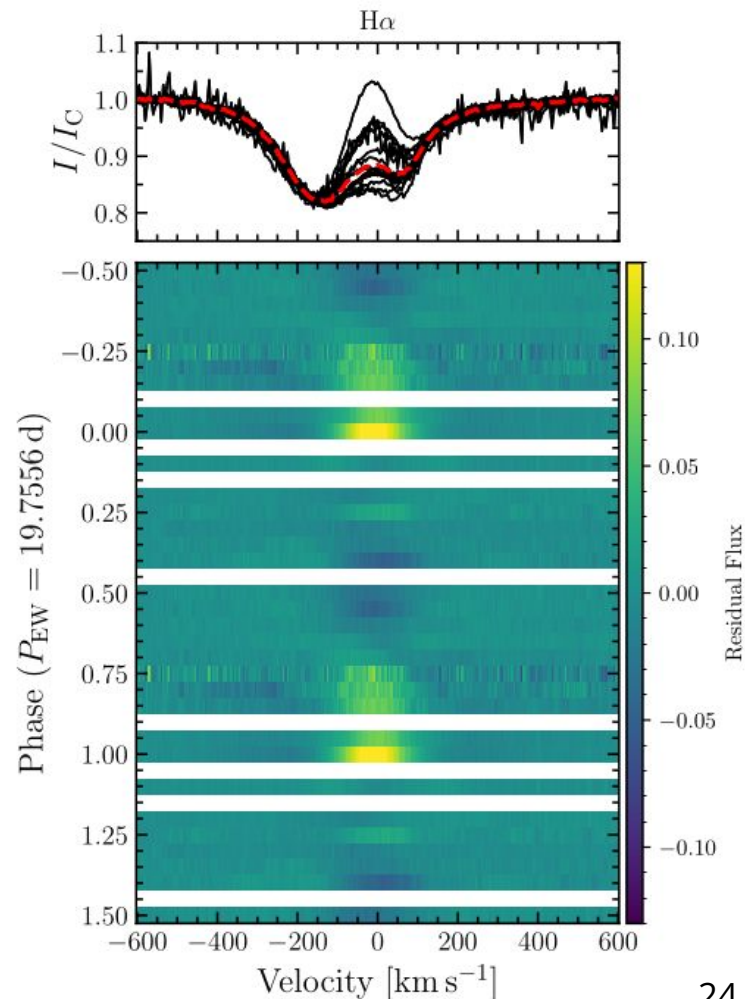


63 Oph: Rotation period of ~ 19.8 days

- Relatively weak Balmer line variability.
- ~ 30 IACOB spectra spanning 15 years.

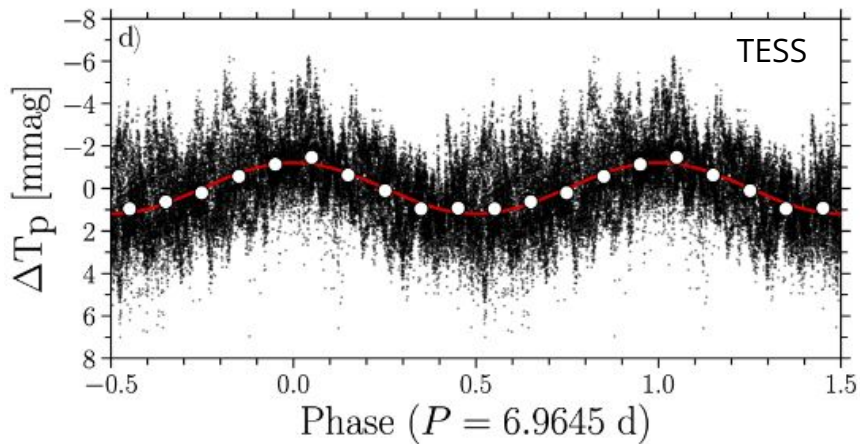


Barron+ 2025



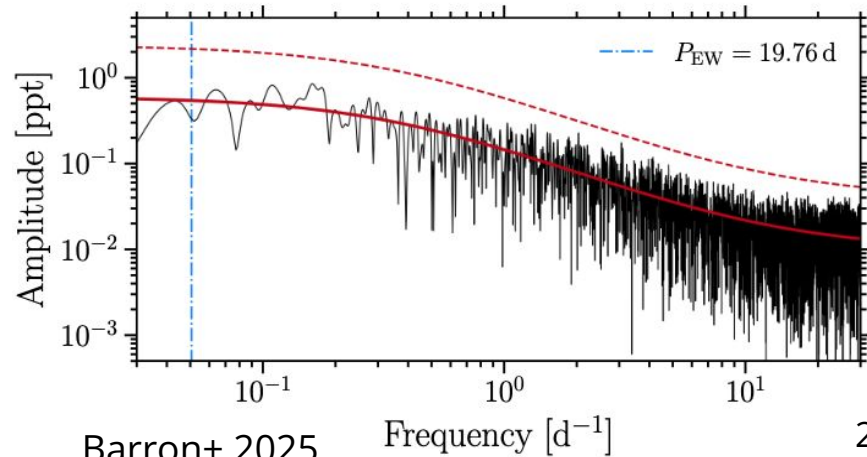
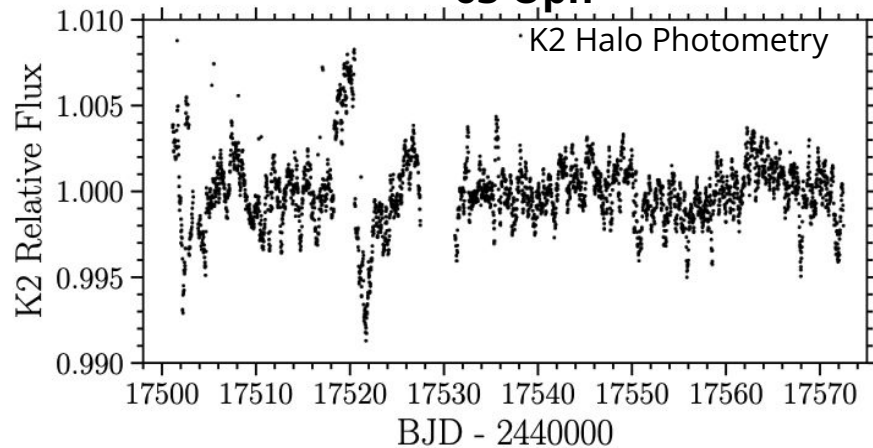
63 Oph: No detected photometric rotational modulation

HD 148937



Frost+ (incl. Wade & Barron), 2024, Science

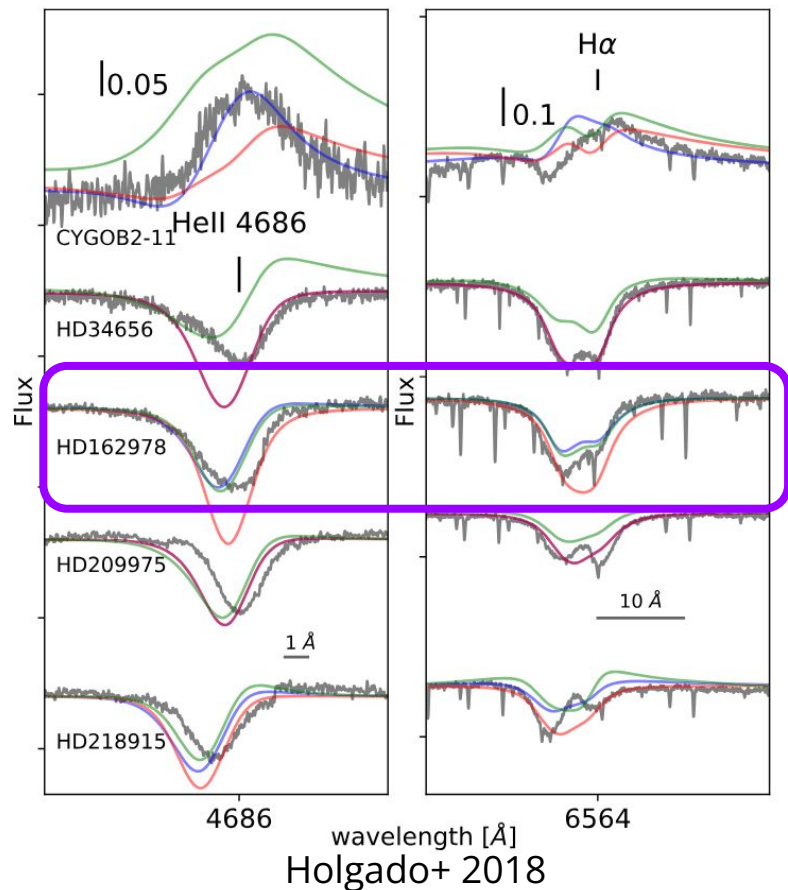
63 Oph



Barron+ 2025

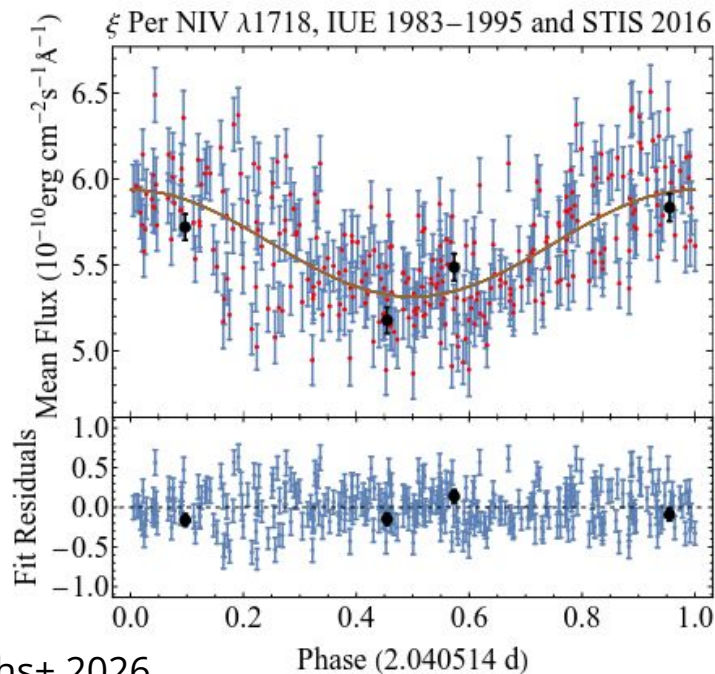
How to identify weakly magnetic O star candidates?

Incompatibility between He II 4686 and H α



Magnetic Proxies

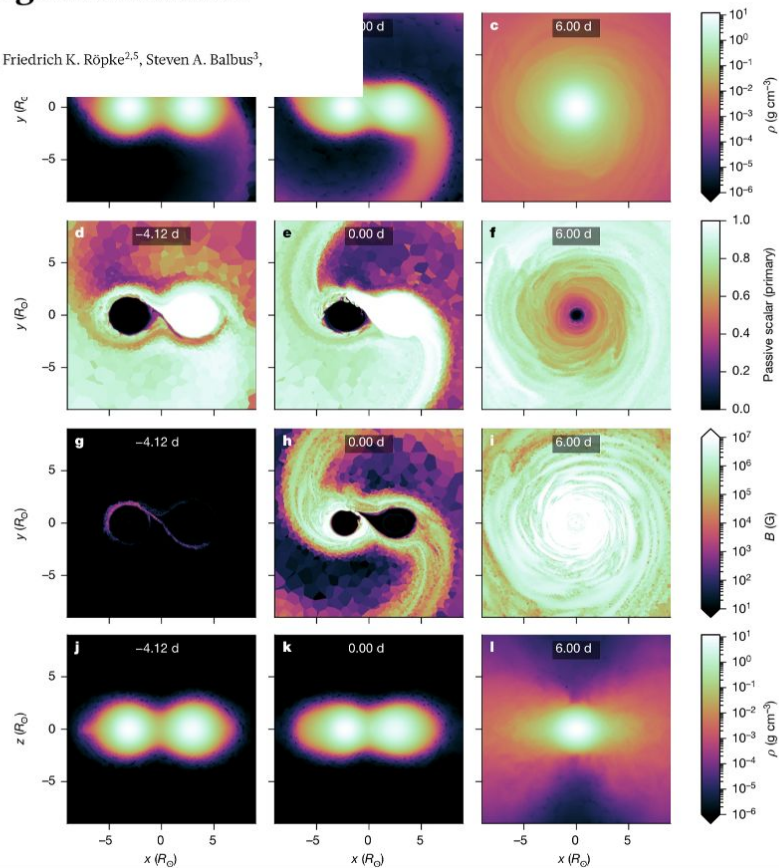
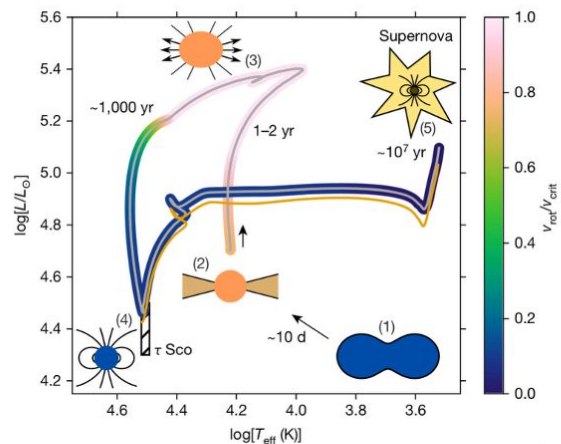
- DACS?
- Periodic photometric and spectroscopic variability?
- Nonthermal radio emission?



3. Magnetic fields and binary evolution

Stellar mergers as the origin of magnetic massive stars

Fabian R. N. Schneider^{1,2,3,7*}, Sebastian T. Ohlmann^{2,4,7*}, Philipp Podsiadlowski³, Friedrich K. Röpké^{2,5}, Steven A. Balbus³, Rüdiger Pakmor⁶ & Volker Springel⁶



A magnetic massive star has experienced a stellar merger

A. J. FROST¹, H. SANA², J. MARIN³, G. WADE⁴, J. BARROD⁵, J.-B. LE BOUQUIN⁶, A. MERIAND⁷, F. R. N. SCHNEIDER⁸, T. SHENAR⁹, J.-J. AUBERT¹⁰

J. V. SMOKER¹¹ +6 authors [Authors Info & Affiliations](#)

SCIENCE • 11 Apr 2024 • Vol. 384, Issue 6692 • pp. 214-217 • DOI:10.1126/science.ade7708

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Editor's summary

Most stellar magnetic fields (including the Sun's) are produced by a dynamo that is generated in interior layers undergoing convection. Massive stars (eight or more solar masses) do not have convective interiors, so it has been unclear why about 7% have magnetic fields. Frost *et al.* observed a binary massive star using interferometry and spectroscopy and found that one component is magnetic and appears to be younger than its companion. They argue that the system originally contained three or more stars and then experienced a merger that produced the magnetic star and made it appear younger. Comparing with theoretical models, the authors reproduced the properties of the system within this merger scenario. —Keith T. Smith

NGC 6154/5, Mark Hanson

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A MASSIVE HELIUM STAR WITH A SUFFICIENTLY STRONG MAGNETIC FIELD TO FORM A MAGNETAR

RESEARCH ARTICLE
STELLAR ASTROPHYSICS
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A massive helium star with a sufficiently strong magnetic field to form a magnetar

TOMER SHENAR, GREGG A. WADE, PABLO MARCHANT, STEFANO BAGNULO, JULIA BODENSTEINER, DOMINIC M. BOWMAN, AVISHAI GILKIS, NORBERT LANGER, ANDRÉ NICOLAS-CHENÉ, J. J. AND SILVIA TOONEN
+6 authors
Authors Info & Affiliations

SCIENCE · 17 Aug 2023 · Vol 381, Issue 6659 · pp. 761-765 · DOI: 10.1126/science.adc3293

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Editor's summary

Magnetars are neutron stars with extremely strong magnetic fields, the origin of which is debated. One possibility is amplification of a magnetic field in the core of the parent star, which produces the neutron star during a supernova explosion. However, such fields have not been observed in stars that are about to explode. Shenar *et al.* spectropolarimetry to identify a high magnetic field on a Wolf-Rayet star, the exposed helium core of a star that has lost its outer layers of hydrogen. The mass of the Wolf-Rayet is high enough that it will produce a neutron star in a supernova, and the field is sufficiently strong to generate a magnetar during core collapse. —Keith T. Smith

(1978). Although sometimes classed as a Wolf-Rayet star, except for the presence of emission lines in its spectrum, HD 45166 differs from true Wolf-Rayet stars in all its physical parameters. Since the spectrum also contains absorption lines characteristic of a late B-type star, Heap and Aller suggest that HD 45166 is a binary system composed of a normal late B star and what they term a “quasi Wolf-Rayet star” (qWR).¹

Van Blerkom+ 78; Hiltner+ 56; Heap & Aller, unpublished

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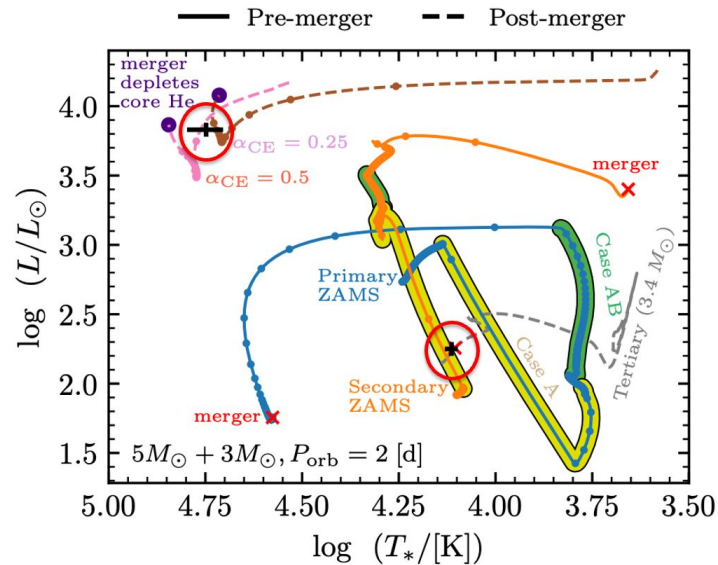
TOMER SHENAR , GREGG A. WADE , PABLO MARCHANT , STEFANO BAGNULO , JULIA BODENSTEINER , DOMINIC M. BOWMAN , AVISHAI GILKIS 

NORBERT LANGER , ANDRÉ NICOLAS-CHÉNE , J.-J. AND SILVIA TOONEN  +6 authors [Authors Info & Affiliations](#)

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RESEARCH ARTICLE | STELLAR ASTROPHYSICS

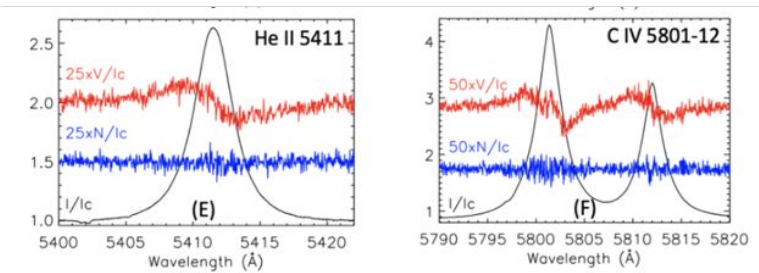
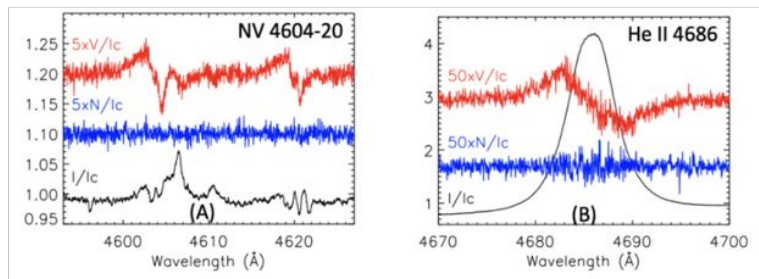
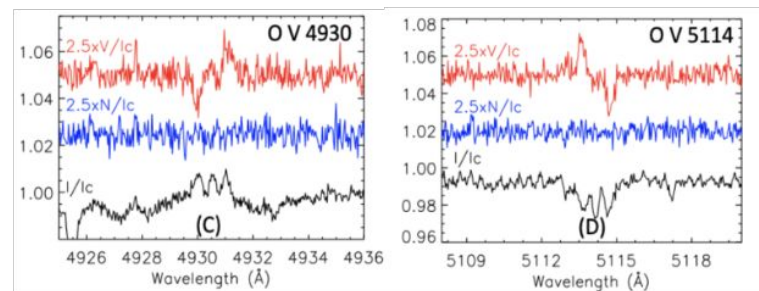
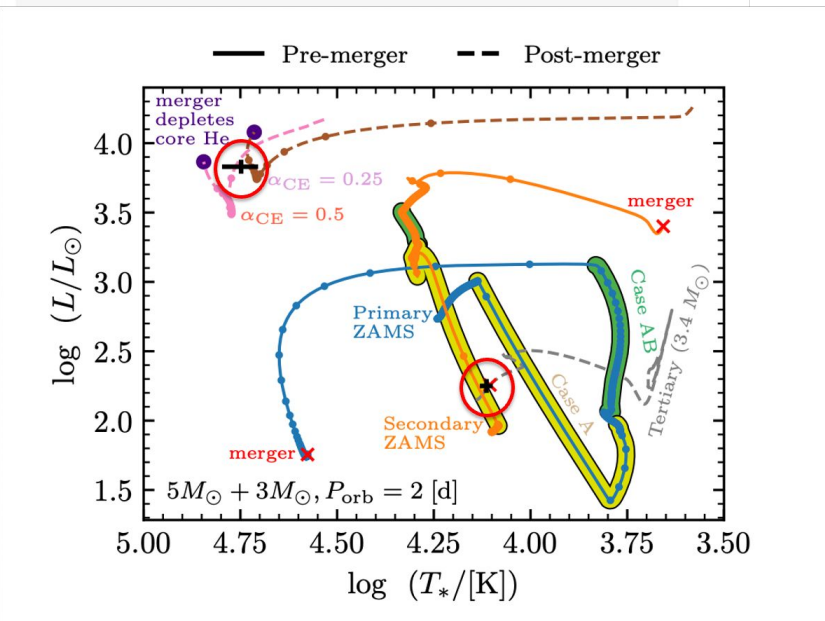
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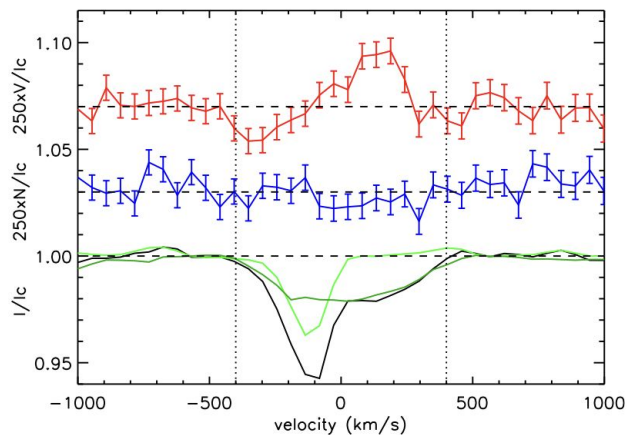
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qWR = magnetic;
B7 V = not magnetic

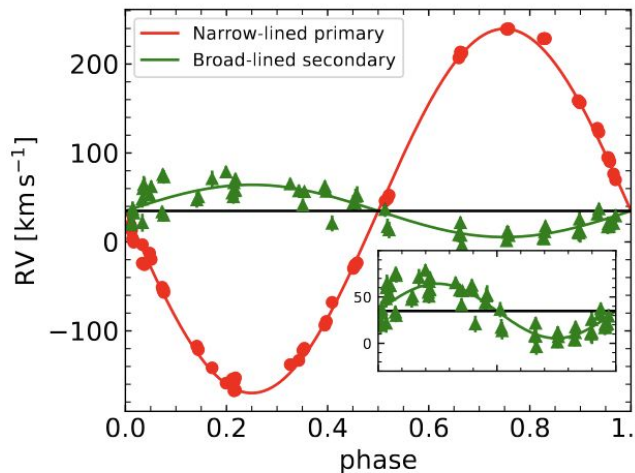
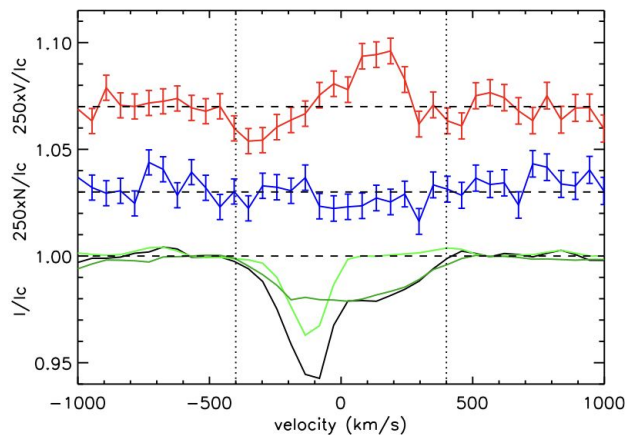
Evidence for the emergence of stellar magnetic fields during rapid mass transfer

Gregg A. Wade^{1,2}, Tomer Shenar³, Pablo Marchant⁴, Michael Abdul-Masih^{5,6}, Matteo Cantiello^{7,8}, Jason Grunhut⁹, Oleg Kochukhov¹⁰, Hugues Sana¹¹, and Selma E. de Mink¹²



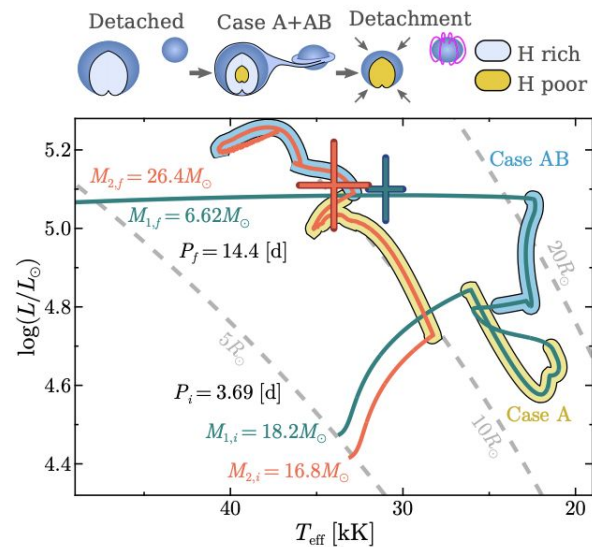
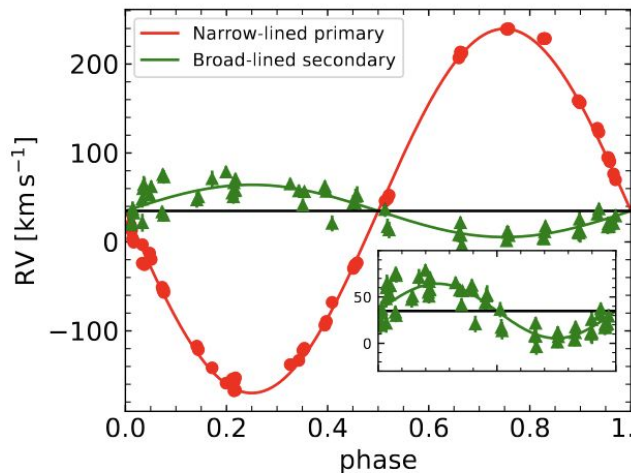
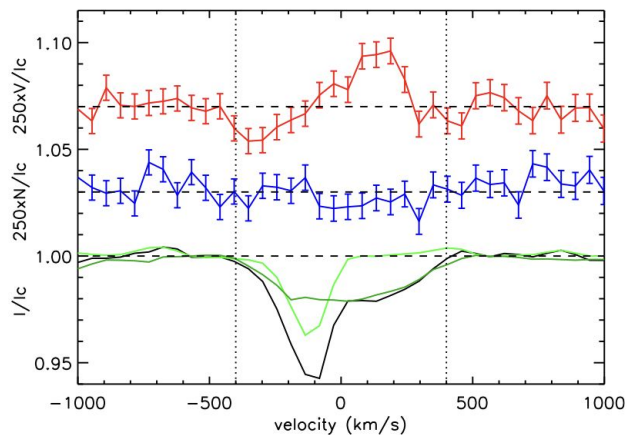
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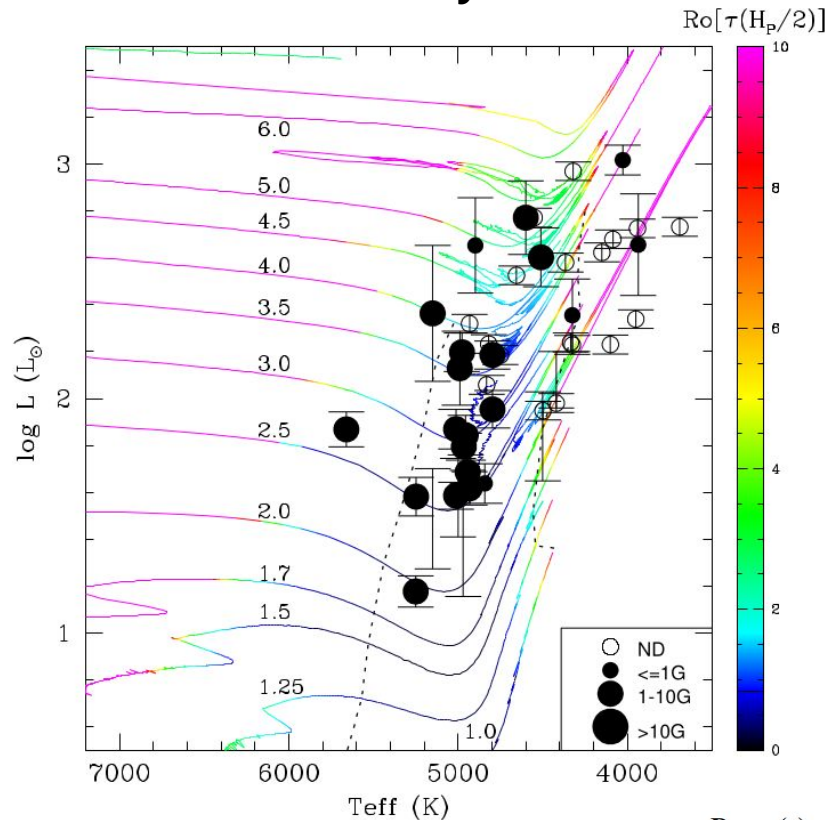
4. Magnetic fields of evolved stars

Magnetic fields in evolved intermediate- and high-mass stars are harder to constrain and are less explored than in their MS counterparts.

1. How do stellar magnetic fields respond to changes in internal structure as their hosts evolve off the MS and become supergiants?
2. How can spectropolarimetry be used to probe dynamics of extended stellar atmospheres?
3. How can magnetic measurements be used to measure P_{rot} and study the rotational evolution of supergiants?

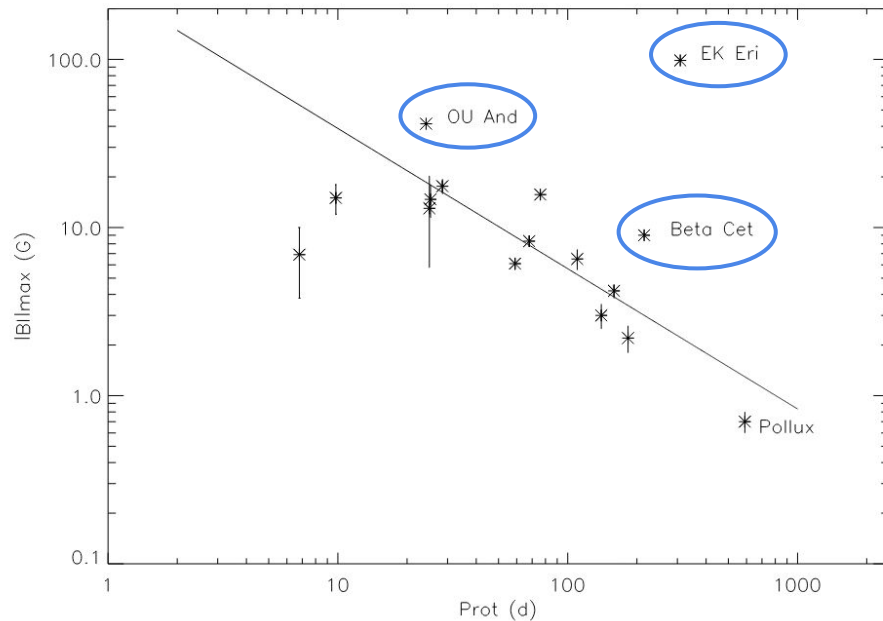
Surface magnetic fields are commonly detected in red giants

Convective Dynamos



Charbonnel+ 2017

Ap Star Fossil Field Descendants?

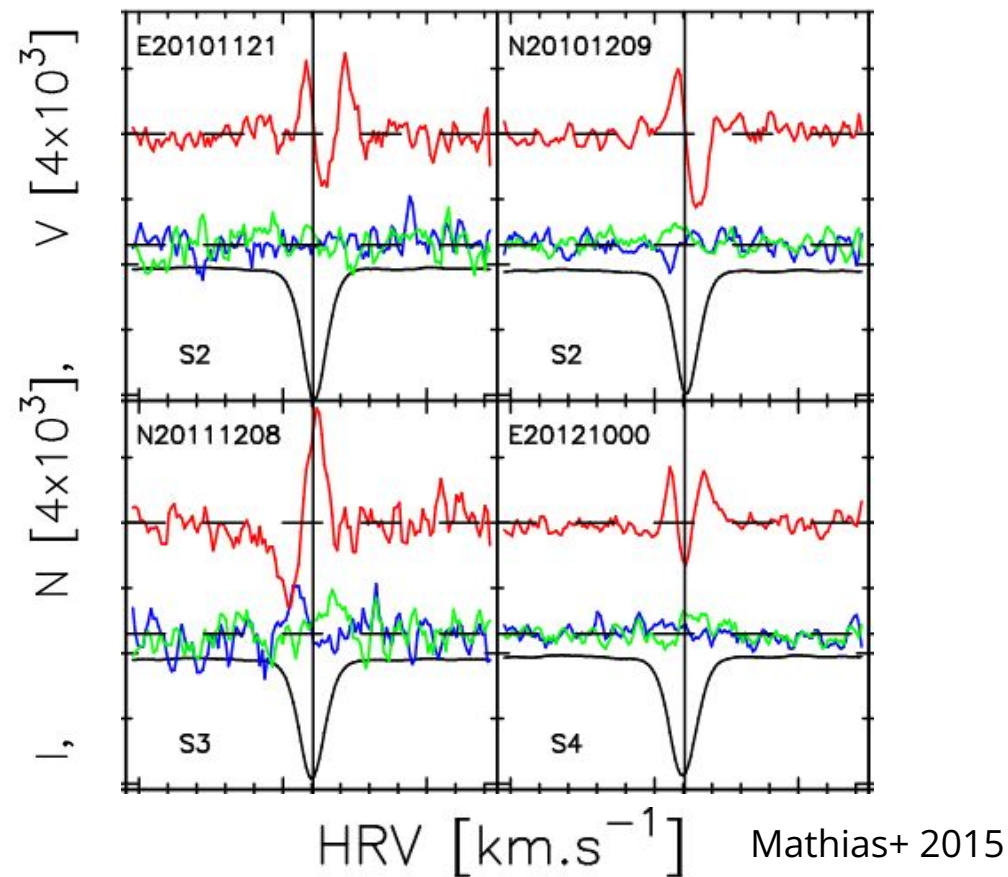


Aurière+ 2015

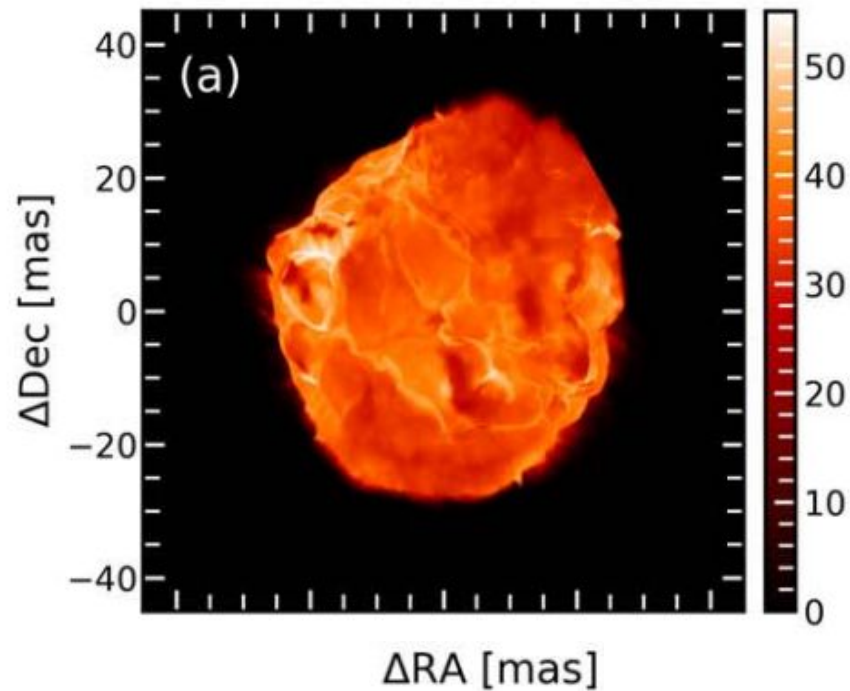
$$B_{surf}(t) = B_{surf}(t_{ZAMS}) \left(\frac{R(t_{ZAMS})}{R(t)} \right)^2.$$

Weak surface magnetic fields are detected in red supergiants

Betelgeuse: Complex Variable Magnetic Signatures

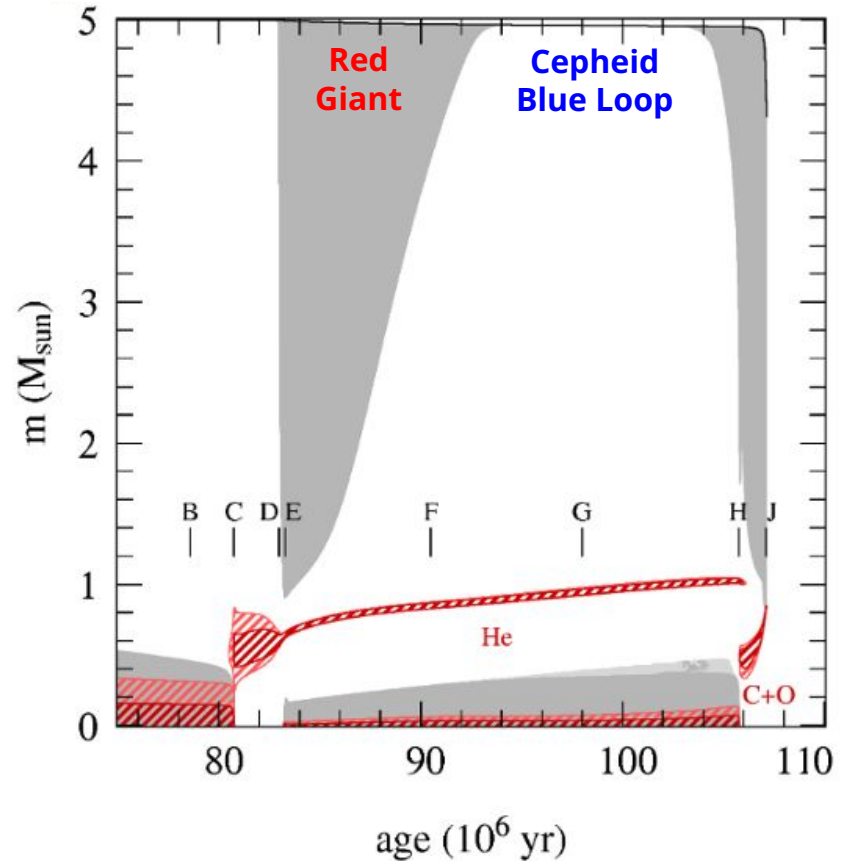
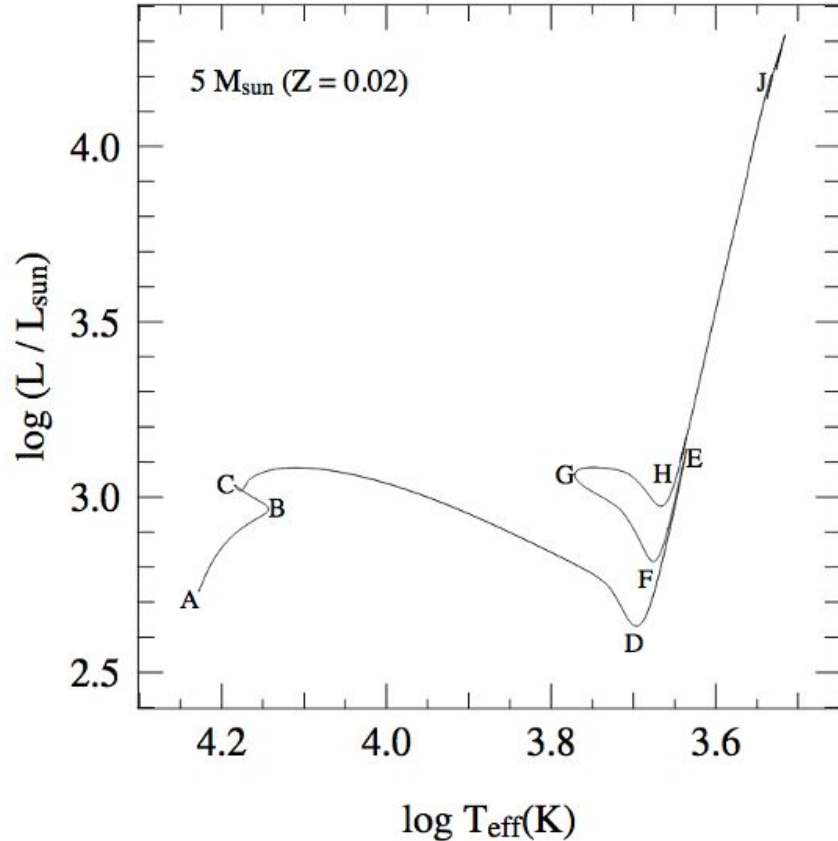


RSG Large Convective Cells



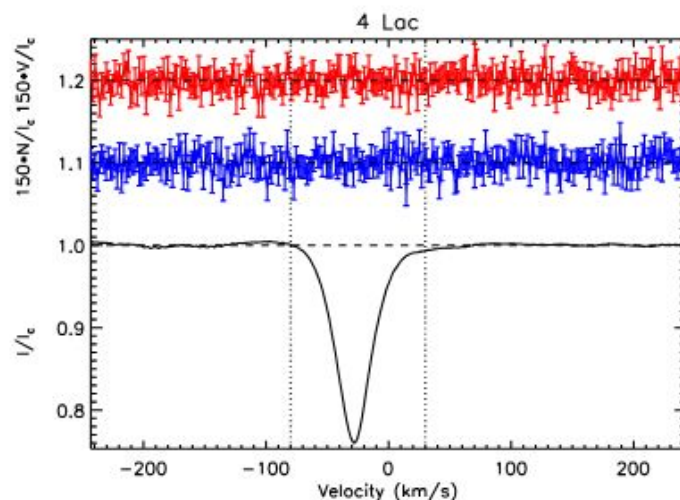
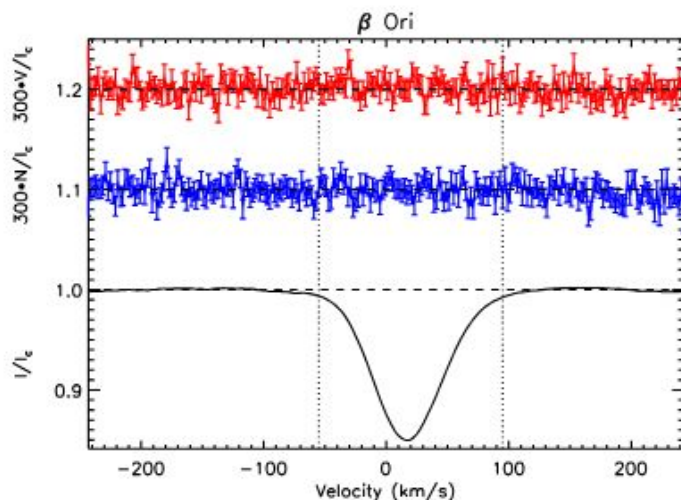
Jing-Ze Ma+ 2024

How do magnetic fields evolve between TAMS and RGB?



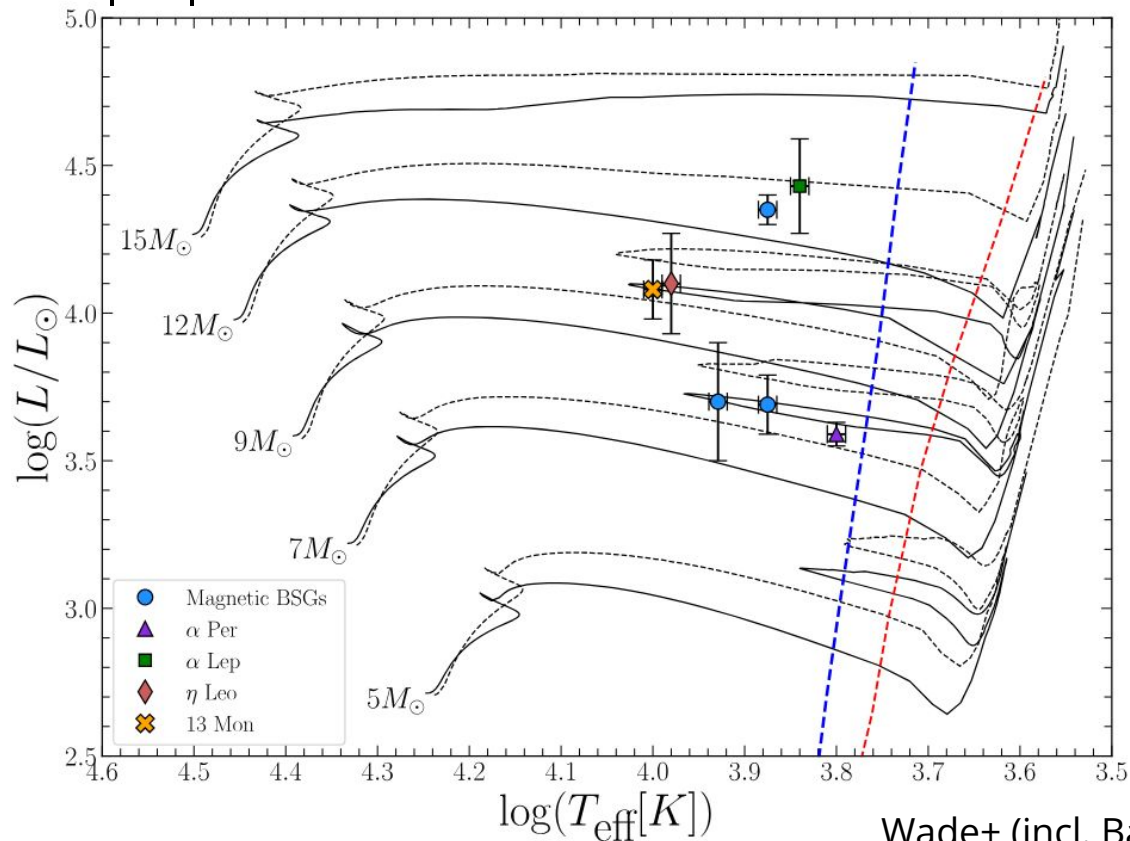
Best current magnetic limits on B-type supergiants

- Shultz+ 2014 obtained ESPaDOnS observations of BA supergiants including Rigel (B8Iae) and 4 Lac (B9Iab). No magnetic detections with $\sim 3 - 10$ G error bars.
- David-Uraz+ 2014, $B_z = -3 \pm 6$ G for eps Ori (B0 Ia).
- LIFE Survey (Martin+ 2018), $B_z = 2 \pm 1$ G for 24 CMa (B3Ia).



Candidate fossil field A and early F type blue supergiants

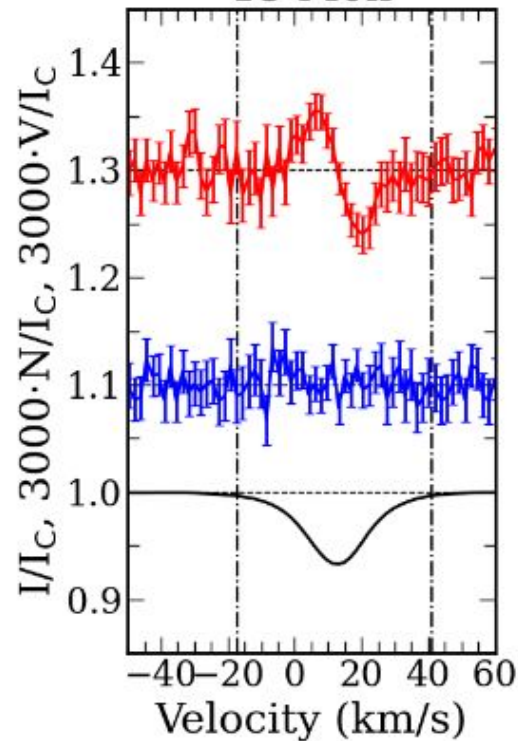
$|B_z| \sim 1 \text{ G}$



Wade+ (incl. Barron), 2025

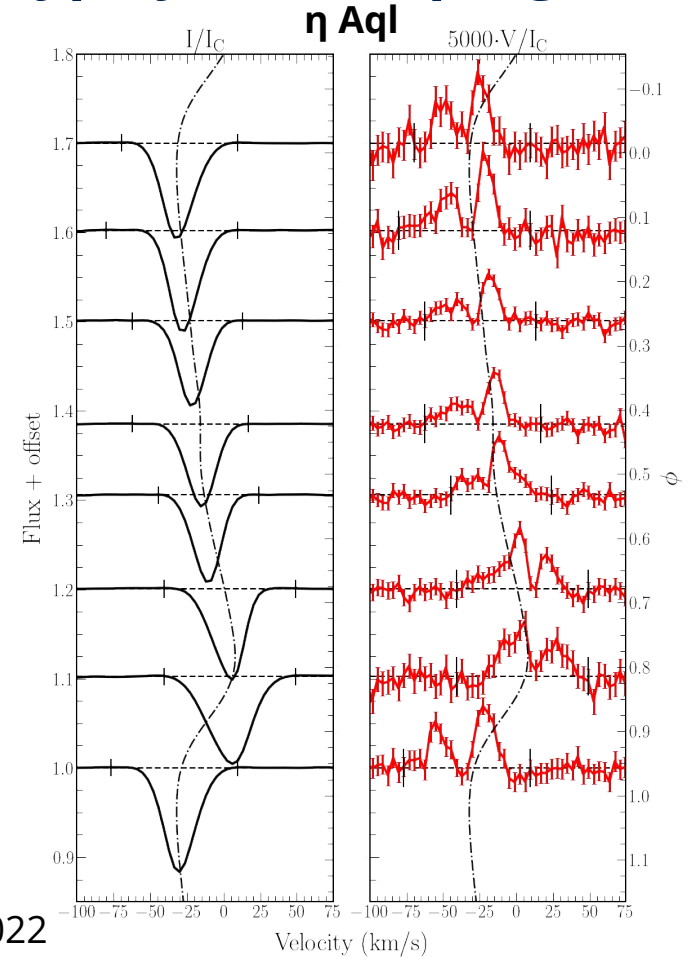
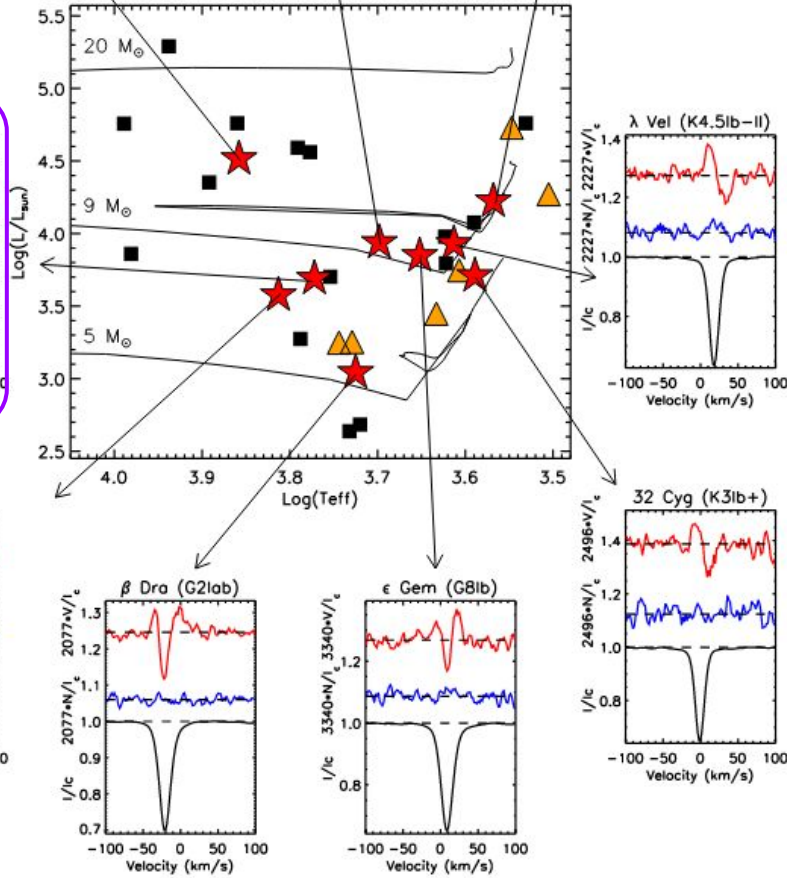
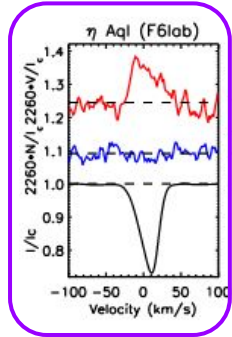
A0Ib

13 Mon



Complex magnetic signatures of late F-type yellow supergiants

Classical
Cepheid

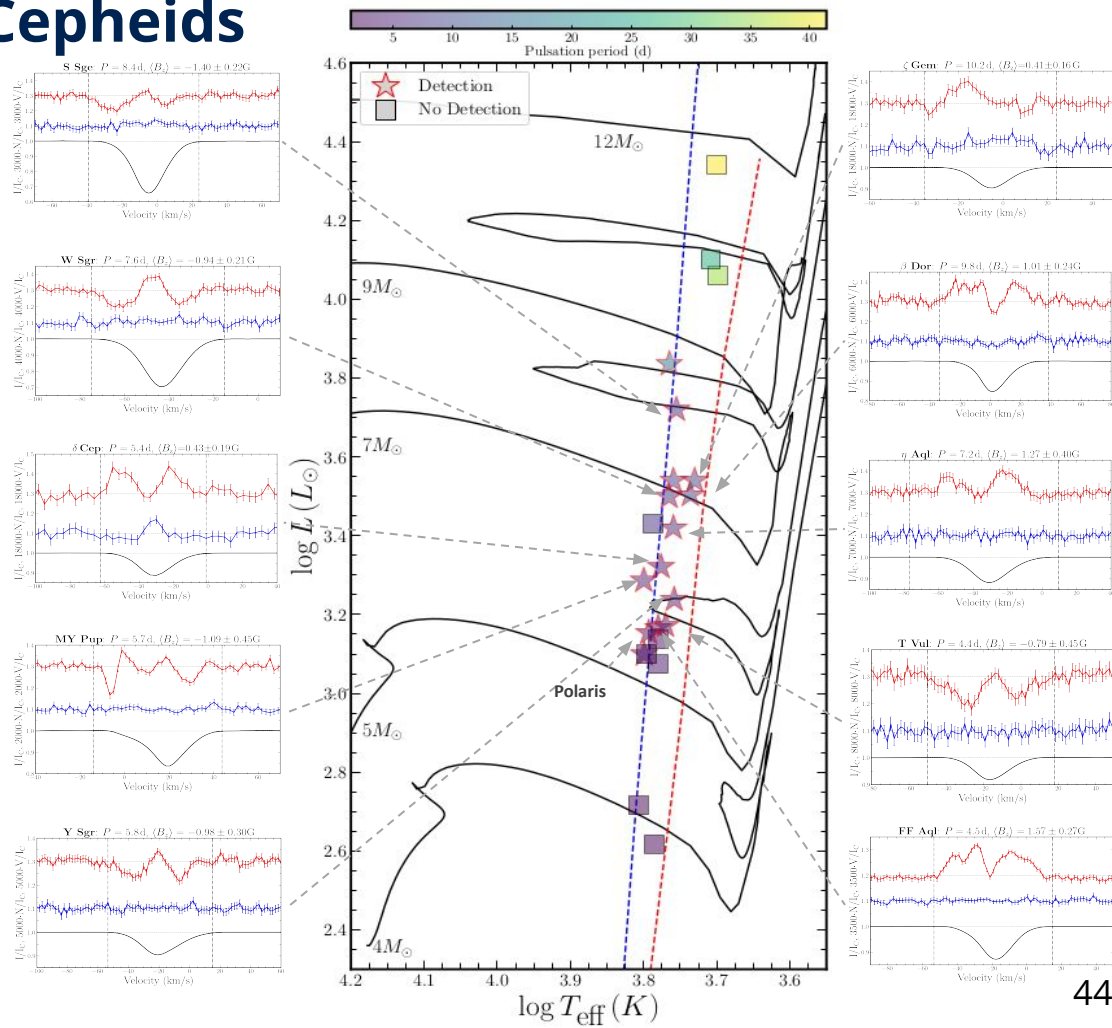


Grunhut+ 2010

Barron+ 2022

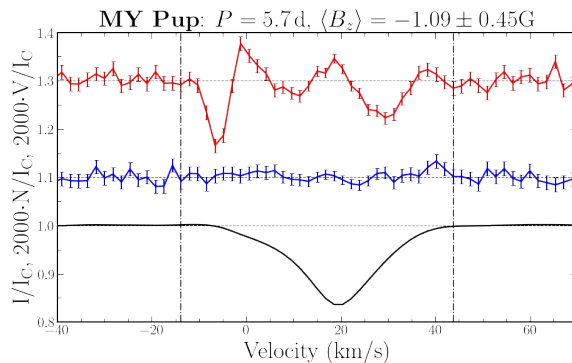
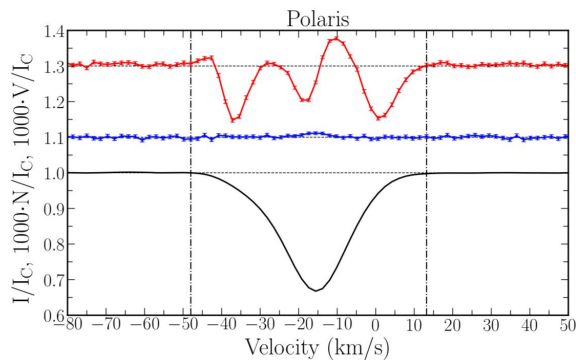
First magnetic survey of Cepheids

- Magnitude limited survey ($V < 6$ mag) of 24 of the brightest CCs using using ESPaDOnS and HARPSpol (Barron+ 2022, in prep).
- Observations obtained near photometric maximum.
- 14 total detections. 10/12 targets detected with ESPaDOnS when $S/N > 3000$ at 500 nm.
- Variety of asymmetric Stokes V morphologies which we attribute to the presence of atmospheric magnetic and velocity gradients.

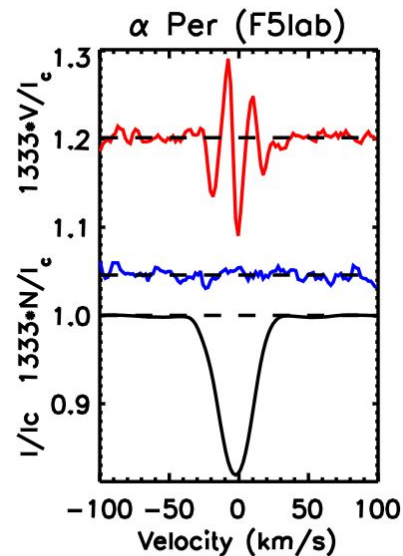
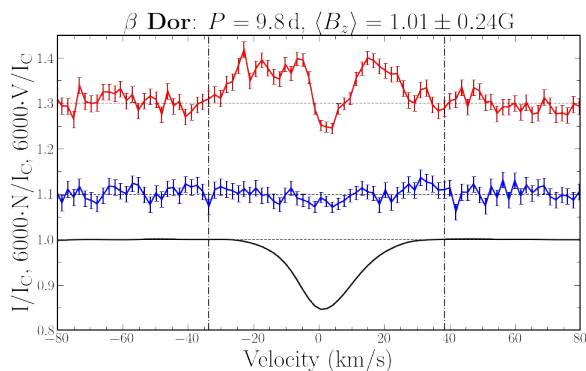
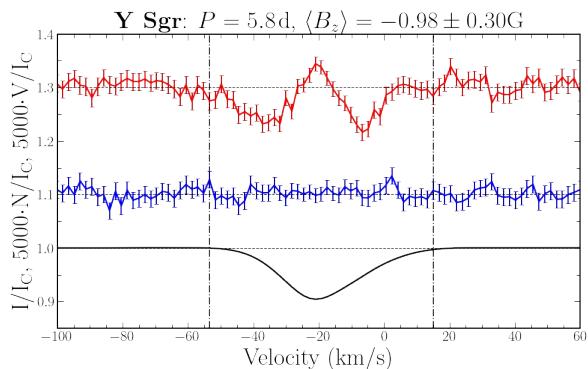


Origin of LSD Stokes V Asymmetry: s-Cepheids vs. Fundamental

Low Amplitude s-Cepheids



Fundamental Cepheids

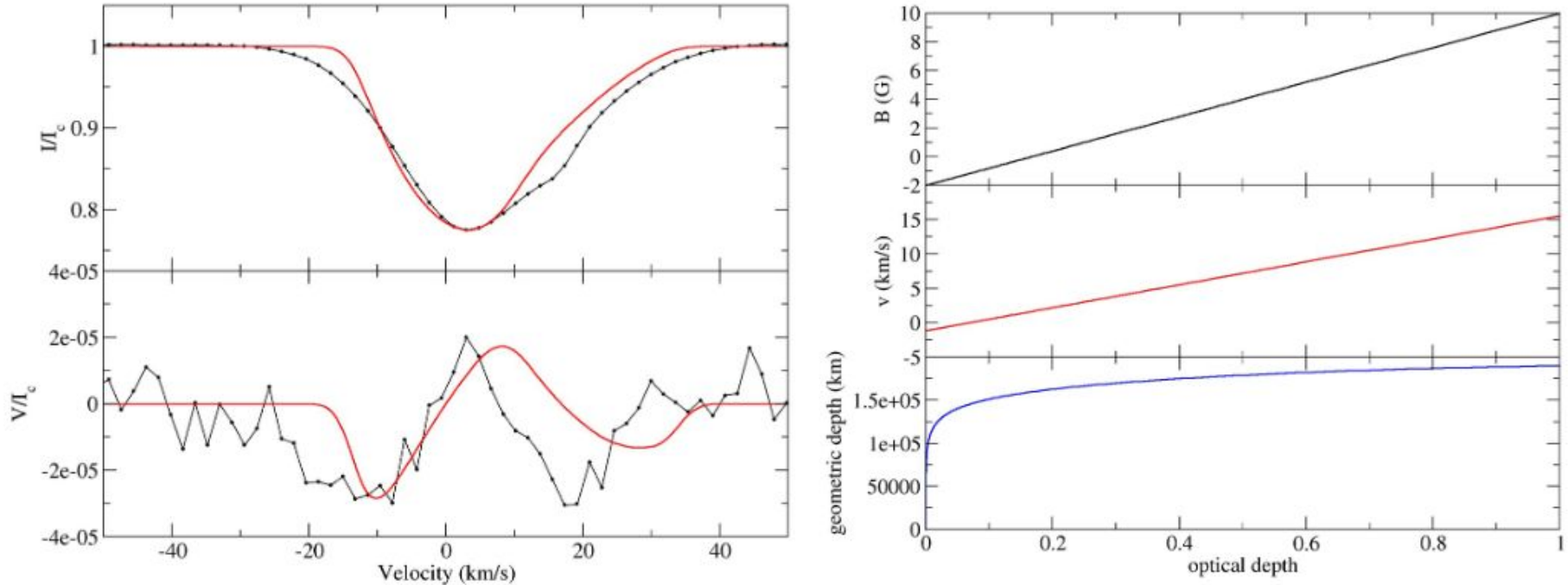


Grunhut et al. 2010

Barron+ 2022, in prep.

Cepheid Stokes V signatures probe atmospheric gradients

Y Sgr - Toy Model



Courtesy of Colin Folsom

Polaris: The North Star is a Triple System

Distance

133 pc

System Components

Polaris (Aa + Ab) + B

Spectral Types

(F8Ib + F6V) + F3V

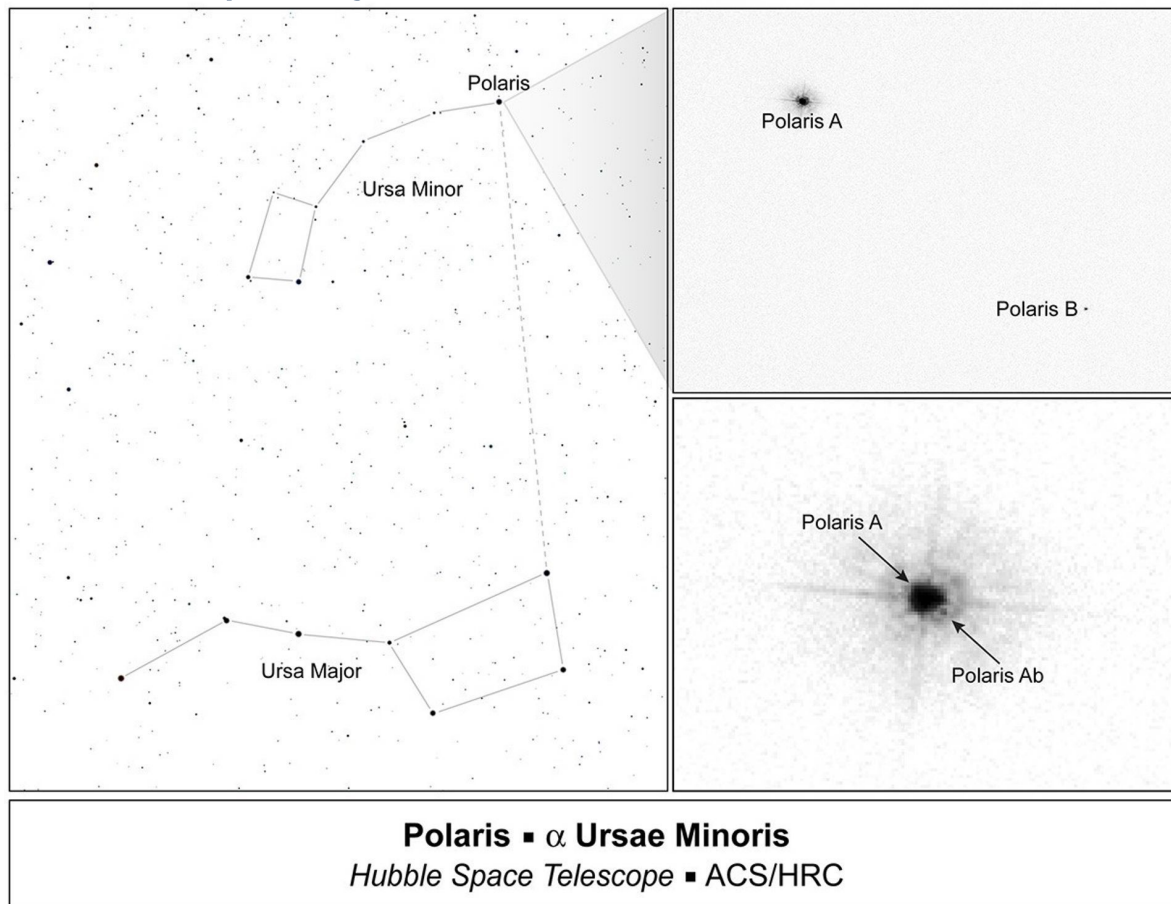
Inner Orbit (Evans+ 2024)

$P_{\text{orb}} = 29.4$ years

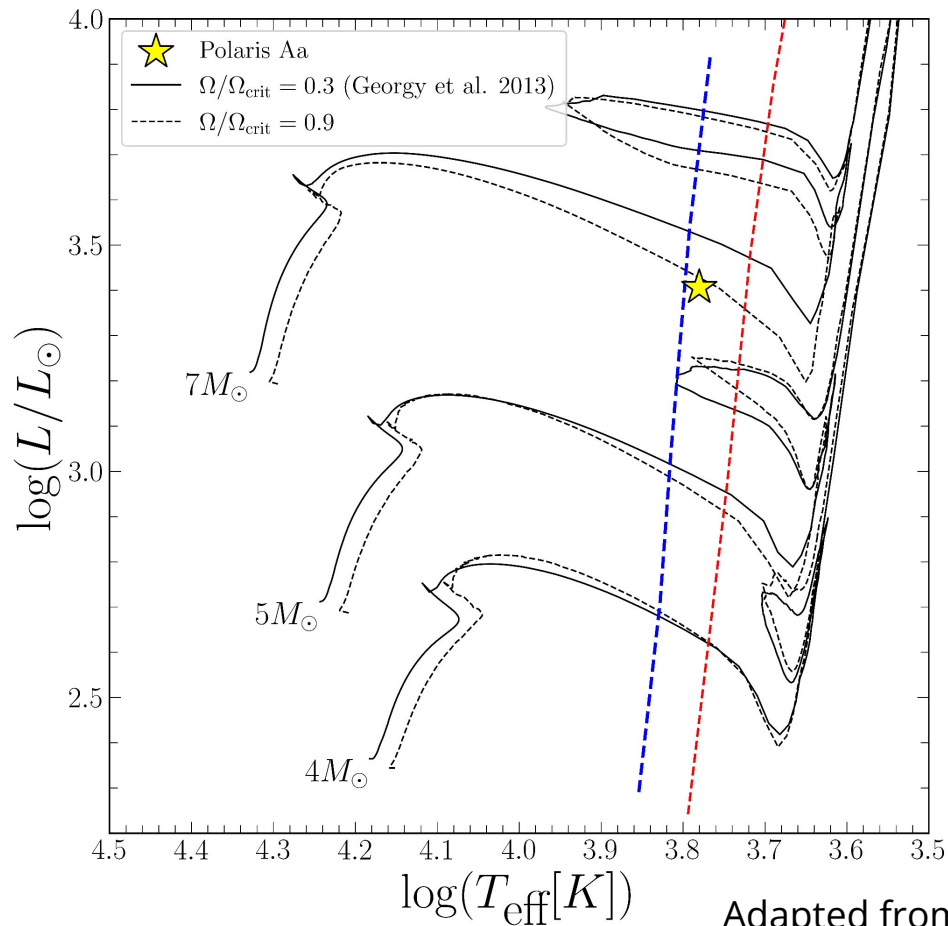
$e = 0.62$

Dynamical Mass

$5.13 \pm 0.28 M_{\odot}$

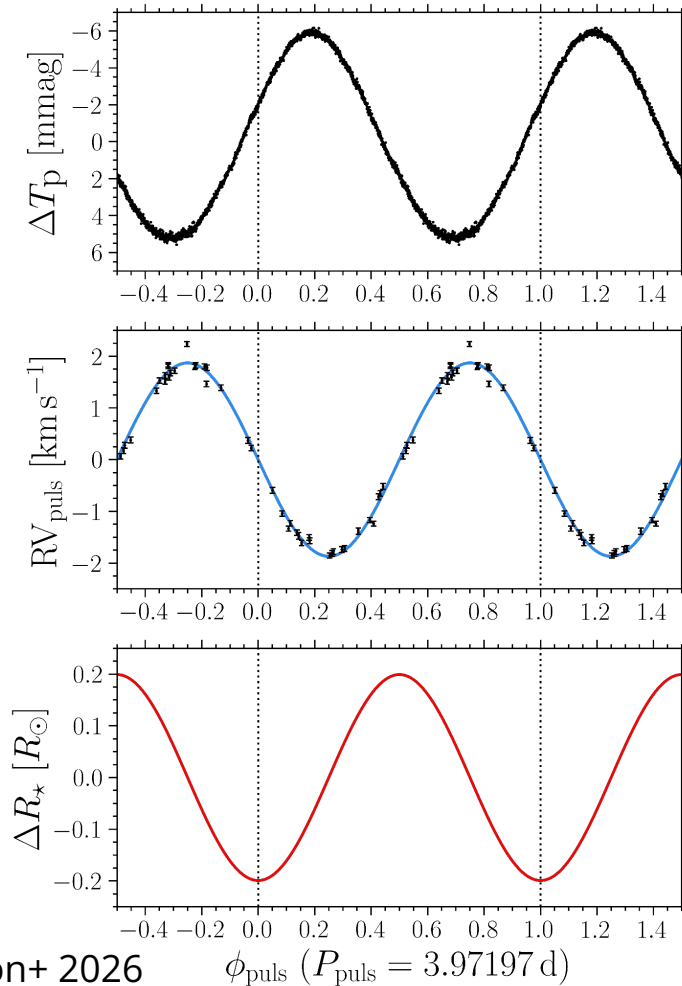


Polaris: An unusual Cepheid



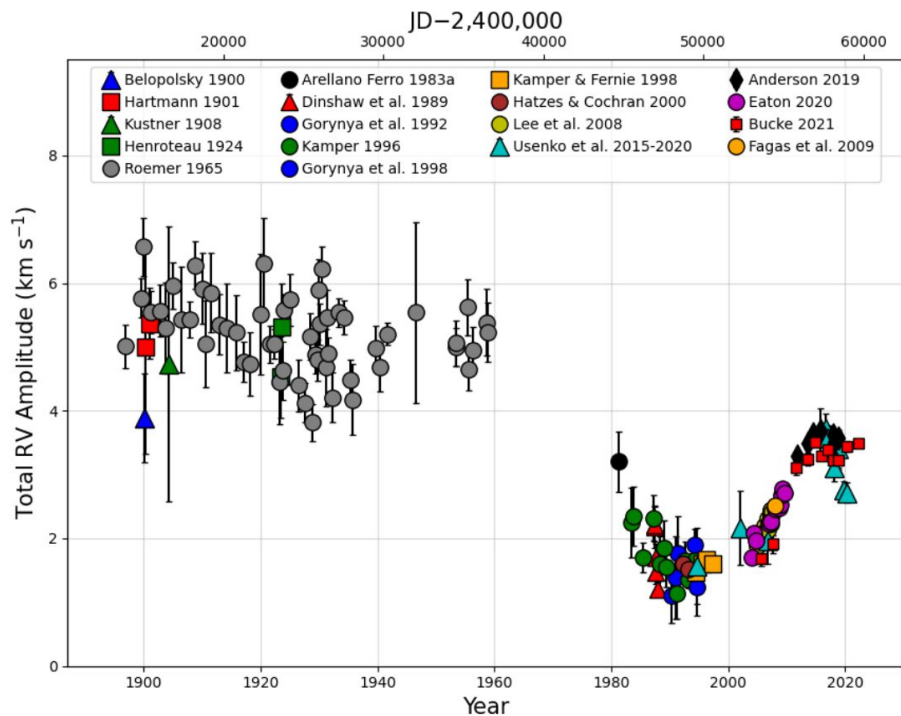
Adapted from Barron+ 2026

Low Amplitude Pulsation



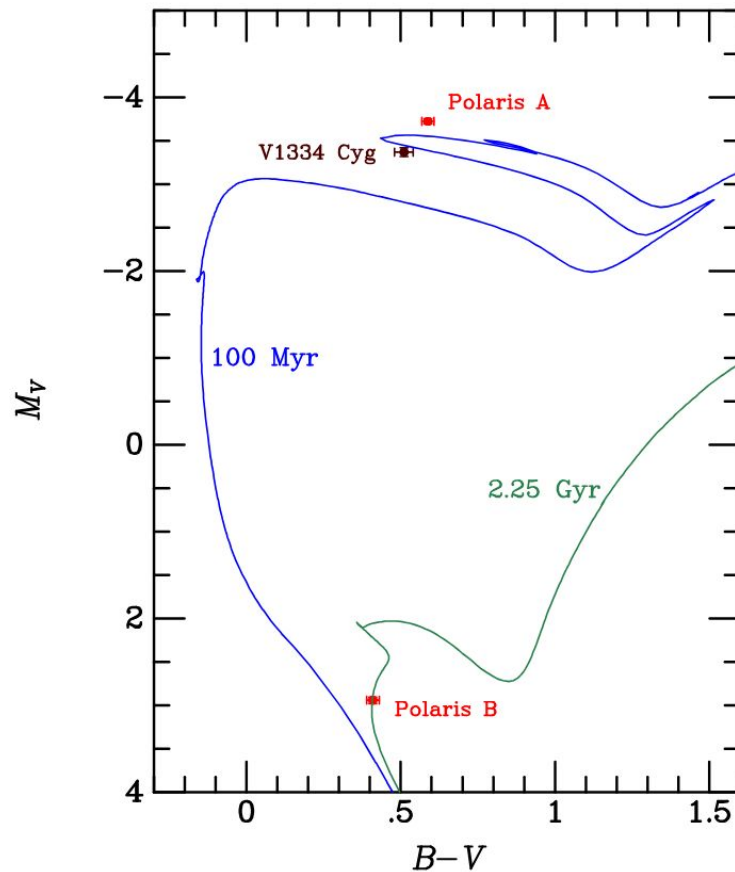
Polaris: Challenges to standard models

Puzzling Variability Changes



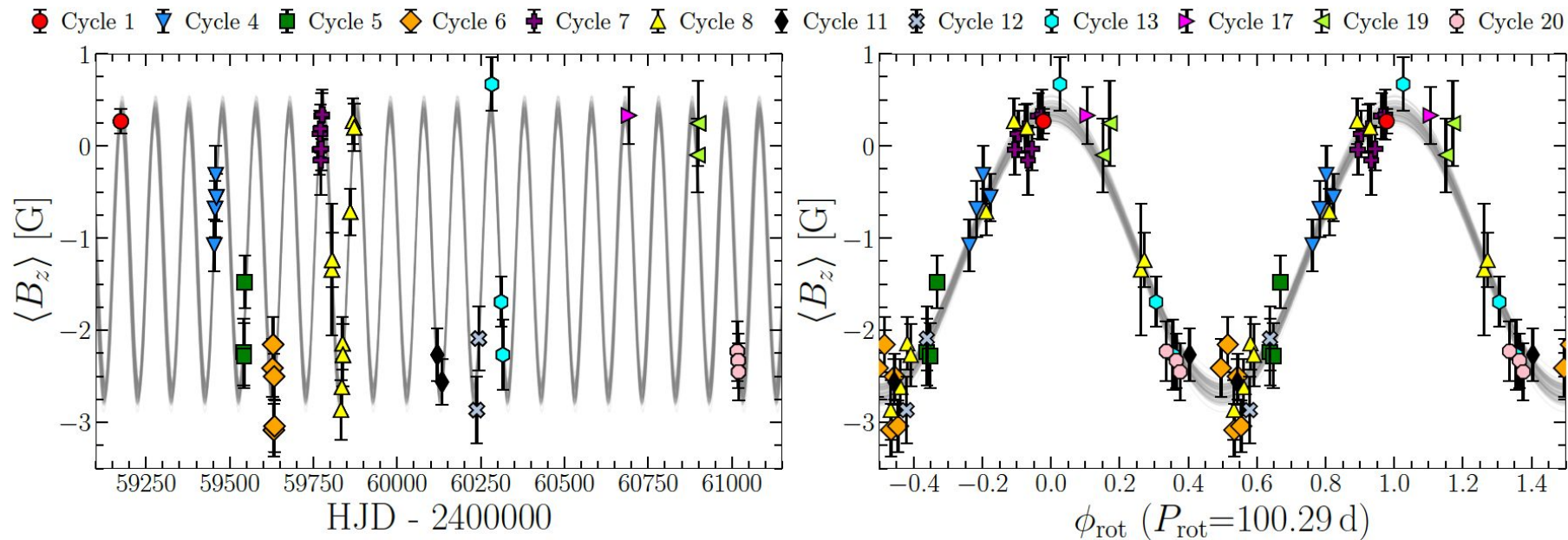
Torres 2021

Merger Hypothesis



Evans+ 2018

Polaris: First direct measurement of P_{rot} for a Cepheid



$$P_{\text{rot}} = 100.29 \pm 0.19 \text{ days}$$

$$v_{\text{eq}} = 23.3 \pm 0.2 \text{ km s}^{-1}$$

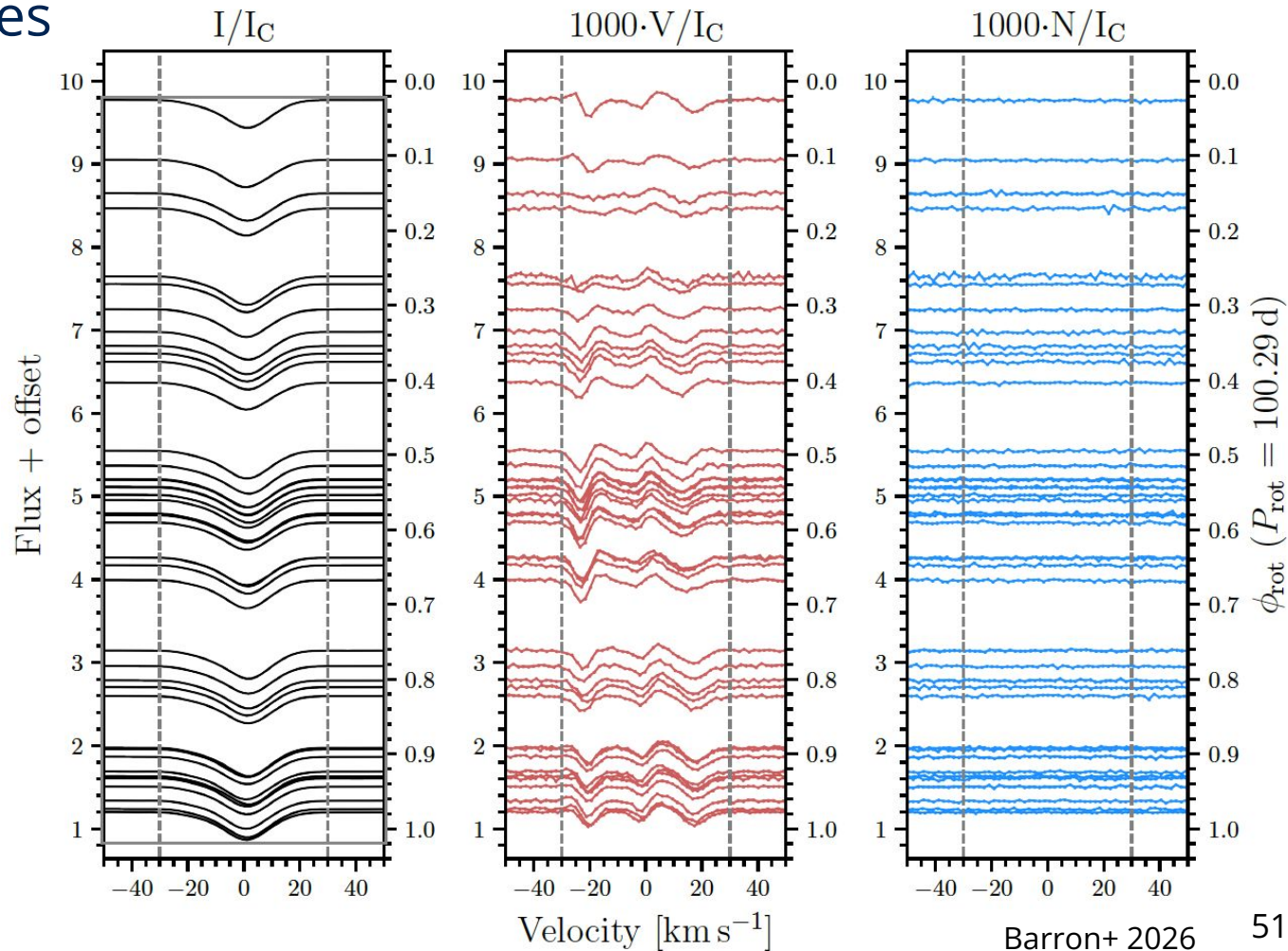
$$v_{\text{eq}} = \frac{2\pi \langle R_{\star} \rangle}{P_{\text{rot}}}$$

Barron, Wade and Folsom 2026

Polaris: LSD Profiles

Field Stability

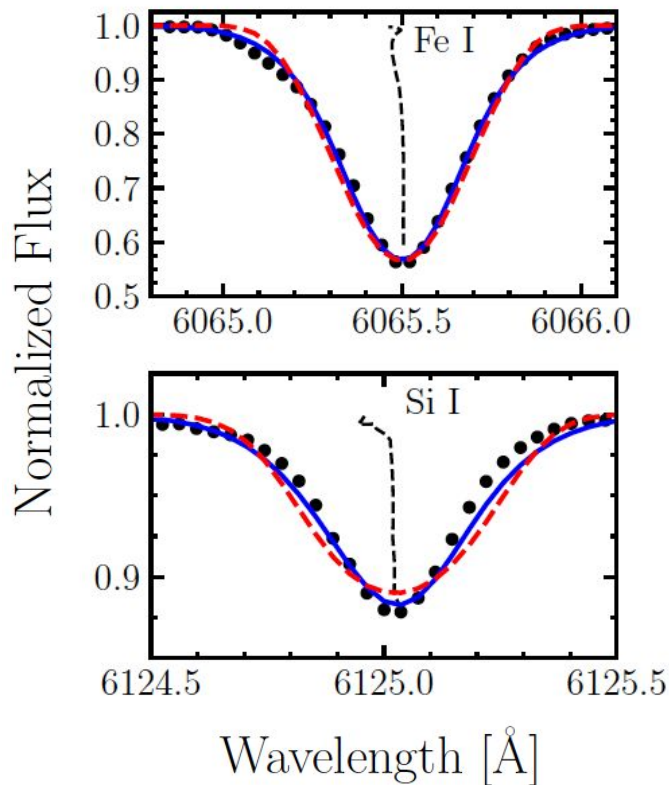
- No clear evidence of magnetic field evolution over 5 years.
- Observations cover 18 rotation and 460 pulsation cycles.



Polaris: High probability of spin-orbit misalignment

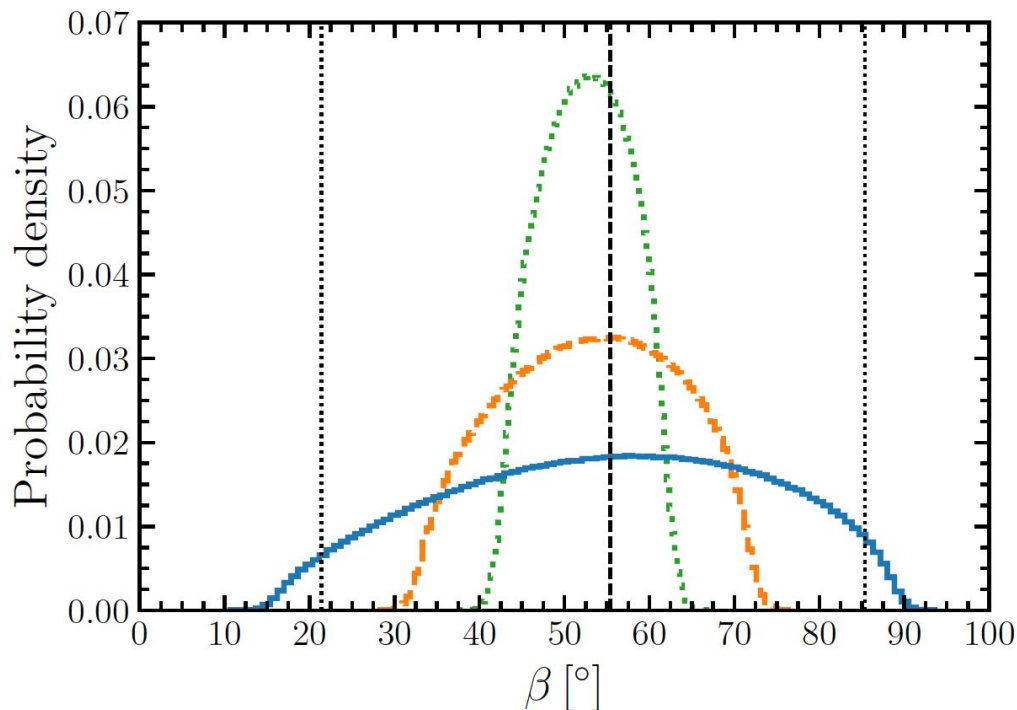
$$v_{\text{eq}} \sin i_{\star} < 13.5 \text{ km s}^{-1} \quad i_{\star} < 37^{\circ}$$

— Best Fit - - - $v_{\text{mac}} = 0 \text{ km s}^{-1}$



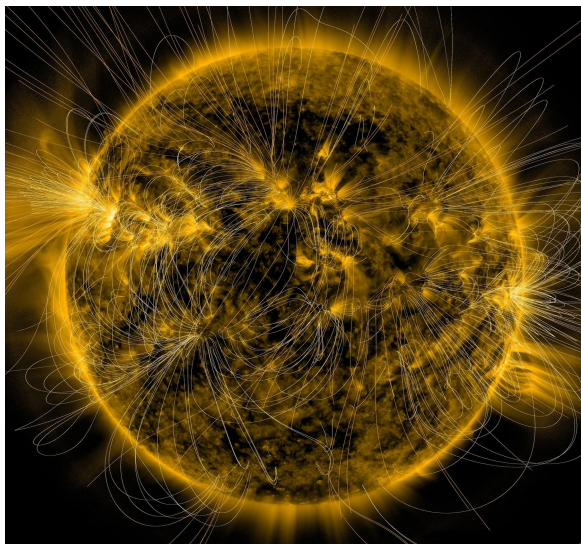
Angle between rotation and orbital axes

□ $i_{\star} < 37^{\circ}$ □ $i_{\star} < 20^{\circ}$ □ $i_{\star} < 10^{\circ}$



What is the origin of Cepheid magnetic fields?

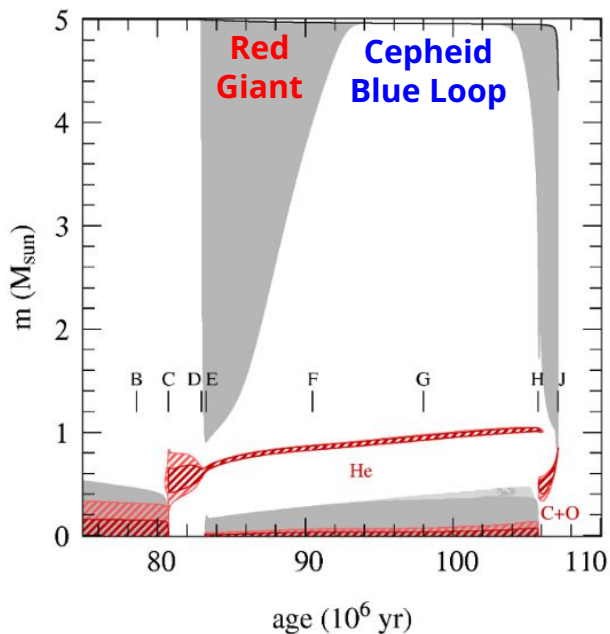
Solar Dynamo



NASA/SDO/AIA/LMSAL

Convective Dynamo?

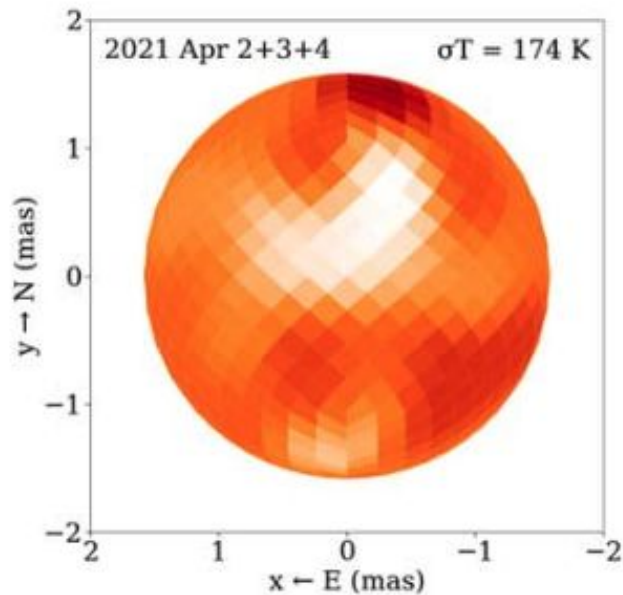
Kippenhahn Diagram



age (10^6 yr)

O.R. Pols

Polaris Interferometric
Surface Reconstruction

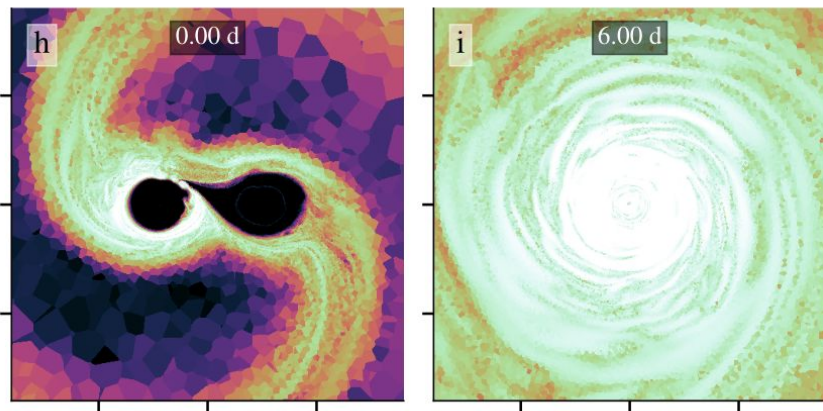


Evans+ 2024

What is the origin of Cepheid magnetic fields?

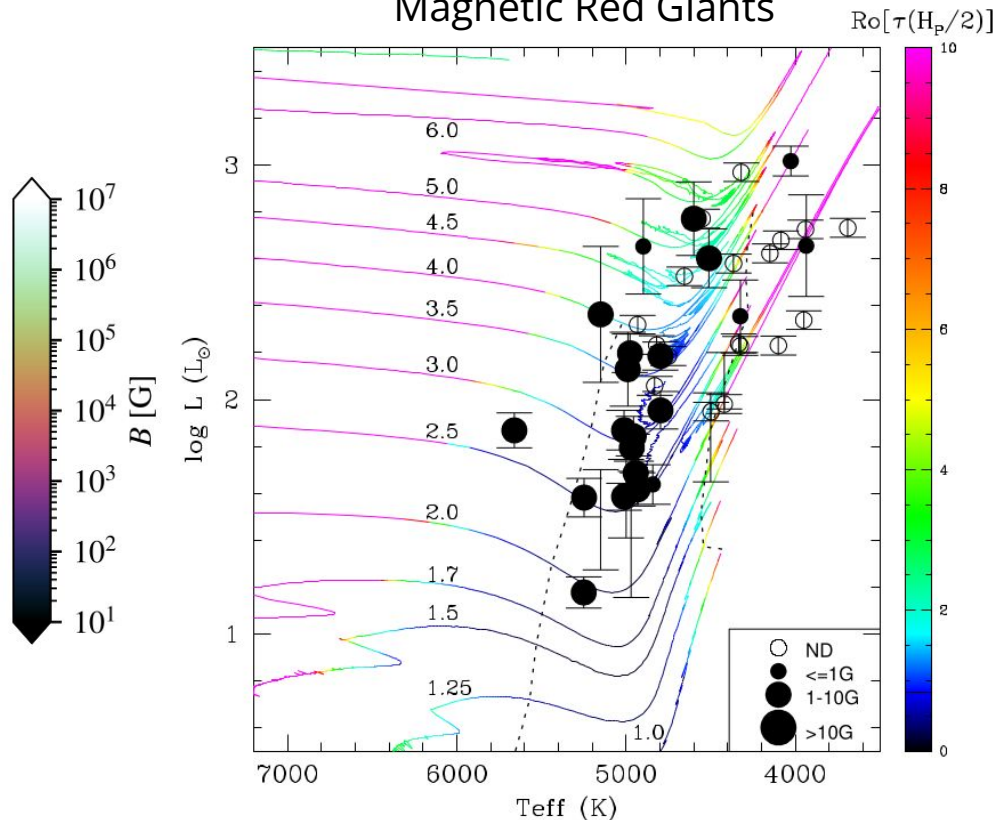
Fossil Field?

Stellar Mergers



Schneider+ 2019

Magnetic Red Giants



Charbonnel+ 2017